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Measurement data recovery for power distribution systems based on physics-guided spatio-temporal graph neural network and online data fusion

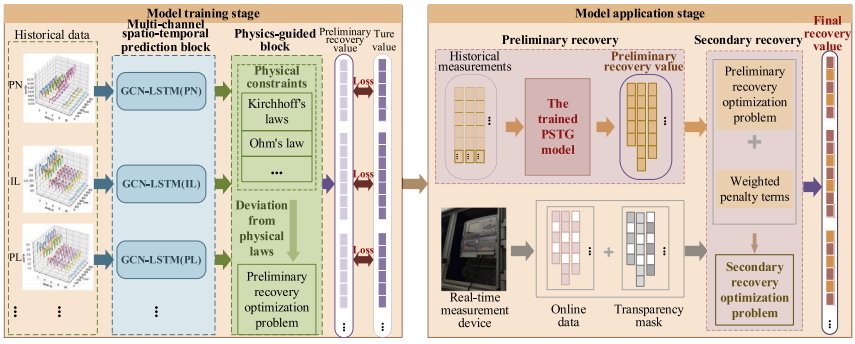
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HIGHLIGHTS

- We propose a coordinated recovery model for heterogeneous missing measurement data in power distribution systems.
- A multi-channel graph construction strategy is proposed.
- Different measurement types couple via physical constraints for mutual calibration.
- An online adaptive data fusion mechanism based on a transparency mask is designed.
- Experiments show significantly improved recovery accuracy.

GRAPHICAL ABSTRACT



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ABSTRACT

Real-time operation control and simulation of power distribution systems are crucial for supporting their platform and intelligent development, requiring reliable online measurement data support. However, resource constraints and cross-system interoperability issues often result in missing or delayed data transmission, demanding efficient recovery solutions. This paper proposes a physics-guided spatio-temporal graph neural network with online data fusion framework for missing data recovery of heterogeneous measurements in power distribution networks, including multiple types such as current, voltage, and power. The framework first establishes the physics-guided spatio-temporal graph neural network-based preliminary recovery model, where a multi-channel spatio-temporal prediction block is employed to extract spatio-temporal features from the measurement data of each channel in parallel, thereby enabling accurate real-time measurement forecasting. Building upon this, a physics-guided module incorporating physical constraints (Ohm's Law and Kirchhoff's Laws) is constructed for post-processing optimization, which couples different types of measurement data through physical constraints to enable mutual calibration and joint recovery, significantly improving recovery accuracy. Subsequently, an online data fusion-based secondary recovery model employs a transparency mask

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mechanism to effectively utilize available online measurements, further refining accuracy. Finally, extensive simulations on the IEEE 33-node distribution system and actual power distribution systems demonstrate that the physics-guided spatio-temporal graph neural network with online data fusion model outperforms conventional methods in recovering missing heterogeneous measurement data, with mean absolute percentage error values reduced by 3.41–1.04% relative to the best existing approaches.

Nomenclature

Abbreviations

PSTG-ODF	Physics-guided spatio-temporal graph neural network with online data fusion
LSTM	Long short-term memory neural network
GCN	Graph convolutional network
IL	Line current magnitude
PL	Line active power
QL	Line reactive power
UN	Node voltage magnitude
PN	Node injected active power
QN	Node injected reactive power

Symbols

$*$	Indicating measurement names, this paper includes IL, PL, QL, UN, PN and QN
N_*	Number of measurement *
P	Number of input features (length of historical time series)
$V^{(*)}$, v_i	Set of measurement points, the i-th measurement point
$E^{(*)}$	Set of edges
$A^{(*)}$,$A_{ij}^{(*)}$	Adjacency matrix for measurement *, its constituent element
z_{ij}	Impedance amplitude of the branch corresponding to edge (v_i, v_j)
K	Number of hidden layers in GCN
S	Total number of nodes in the power distribution system
Ψ_m,Θ_m	Set of predecessor nodes and set of successor nodes for node m
λ	Threshold for constraining measurement value fluctuations
G_{mj}, B_{mj}	Conductance and susceptance of branch (m, j)
R_{mj}, X_{mj}	Resistance and reactance of branch (m, j)
T_*	Total number of missing time steps for measurement *
τ	Duration of missing data, quantified by the number of affected time steps
η	Missing rate
$D^{(*)}$	Raw data set for measurement *
$X^{(*)}$,$X_T^{(*)}$	Input feature matrix and features of the T-th time step
I	Identity matrix
$Y_T^{(*)}$	Real measurement data for the T-th time step
$\widehat{Y}_T^{(*)}$,$\overline{Y}_T^{(*)}$,$\widetilde{Y}_T^{(*)}$	The predicted, preliminary recovery, and final recovery data for the T-th time step
L^{sym}	Symmetrically normalized Laplacian
$\widetilde{D}$	Degree matrix
$H^{(k)}$	Output of the k-th GCN layer (input of the $(k+1)$-th GCN layer)
$W^{(k)}$	Learnable weight matrix for the k-th GCN layer
x_t	Input of LSTM
f_t,i_t,o_t	Output of LSTM forget gate, input gate, and output gate
$\widetilde{C}_t$,C_t,h_t	Candidate state, cell state and hidden state in LSTM
W_f, W_i, W_c, W_o	Learnable weight matrices in LSTM
b_f, b_i, b_c, b_o	Bias vectors in LSTM
W_y, b_y	Weight matric and bias vector of fully connected layer
m	Node in the power distribution system (distinguished from measurement point v_i)
P_m, Q_m, U_m	Injected active power, reactive power and voltage magnitude measurement at node m
I'_{im},I''_{im},I_m	Current measurement at the sending and receiving end of branch (i, m), and at the high-voltage side of the distribution transformer for node m
P'_{im},Q'_{im},P''_{im},Q''_{im}	Active and reactive power measurement at the sending and receiving end of branch (i, m) in the power distribution system
$\xi_T^{(*)}$	Online measurements (with missing data)
$M_T^{(*)}$	Transparency mask
ϑ	Transparency
$\theta_k^{(*)}$	Secondary optimization solution vector at the k-th iteration
$\Gamma_k^{(*)}$	Total penalty term at the k-th iteration
$\Gamma_{k,1}^{(*)}$,$\Gamma_{k,2}^{(*)}$	Weighted penalty term for online measurements and preliminary recovery data at the k-th iteration
$f(\cdot)$	Mapping function of spatio-temporal prediction block
$\text{ELU}(\cdot)$	Activation function of GCN
$\sigma(\cdot)$	Activation function of LSTM (sigmoid function)
$\tanh(\cdot)$	Hyperbolic tangent function
$\mathbb{F}(\cdot)$	Basic belief assignment function

1. Introduction

With the acceleration of global energy transition, power distribution systems worldwide are facing common challenges arising from the widespread integration of new types of equipment such as renewable energy generation and electric vehicles, as well as new business models like virtual power plants. Against this backdrop, power distribution systems are undergoing rapid digital and intelligent transformation. Accurate online measurement data serves as a crucial foundation for this transformation, directly affecting the operational efficiency and security of core functions such as renewable energy integration, coordinated electric vehicle charging, and virtual power plant cooperative control [1]. However, in practice, factors such as device operational conditions and cross-system interoperability often lead to missing or delayed data transmission (e.g., current, power), severely impairing online intelligent perception and operational simulation. Moreover, constrained by investment limitations, the completeness of measurement device configurations varies across different scenarios, further compromising the supporting role of measurement data. Therefore, the online recovery of missing measurement data has become a critical problem to be solved for the transformation and upgrading of power distribution systems, and it also represents a research hotspot of common interest in both international academia and industry.

The measurement data in power distribution systems inherently exhibits complex spatio-temporal correlations [2]. Temporally, the power distribution system contains a large volume of historical measurement data, which exhibit nonlinear dynamics and periodicity due to

environmental variations and load patterns. Spatially, topological connectivity induces strong correlations between nodes. Furthermore, the power distribution system comprises a vast number of measurement devices, which collect diverse types of data such as current, voltage, and power. These measurements are governed by physical laws, including Ohm's law and Kirchhoff's laws, resulting in strong physical correlations.

Early approaches to measurement data recovery predominantly relied on mathematical-statistical methods. In [3], missing data are reconstructed using interpolation techniques. In [4], a streaming principal component analysis (PCA) and subspace tracking algorithm is proposed for data recovery. In [5], the recovery of missing measurements is formulated as a low-rank matrix completion problem. In [6], a regularized low-rank tensor completion (LRTC) method is proposed for the recovery of synchrophasor measurement data. In [7], on the basis of the low-rank matrix framework, nuclear norm minimization and bayesian estimation methods are employed to incorporate prior knowledge of the power system in the form of second-order statistics. However, these methods are developed primarily based on either temporal correlations or spatial correlations among the data, without fully accounting for the complex spatio-temporal coupling inherent in measurement data or the underlying physical interdependencies among heterogeneous measurements.

In recent years, with the continuous advancement of machine learning, researchers have begun to integrate it into data recovery applications. In [8], missing information in energy/distribution management systems is addressed by autoencoders. In [9], matrix factorization is incorporated into a generative adversarial network (GAN) for data recovery. Furthermore, long short-term memory neural network (LSTM) [10], which are adept at handling time-series data, and graph convolutional network (GCN) [11], designed for non-euclidean structured data, have also been applied for missing data restoration. In [12], the whale algorithm-optimized convolutional neural network (CNN)-LSTM model is applied to false data injection attack detection and data recovery, realizing high-accuracy data recovery in attack scenarios. These methods fully leverage the extensive historical measurement data available from the real physical system, thereby enhancing data recovery accuracy and adaptability to some extent. However, they still fail to adequately integrate the spatio-temporal correlations within the measurement data, limiting the overall recovery performance [13].

Building on this foundation, researchers have further proposed measurement recovery methods based on spatio-temporal graph neural network (STGNN). By jointly modeling spatio-temporal relationships through graph structures, these methods effectively capture nonlinear relationships and spatio-temporal correlations, demonstrating superior potential for distribution system data recovery. A graph attention convolutional network for data imputation is proposed in [14], a spatio-temporal kriging framework combining spatial aggregation with temporal convolutional networks is proposed in [15], and a graph neural network-based method for multivariate time series imputation is proposed in [16]. In [17], the hankel matrix of spatio-temporally synchronized phasor data is lifted to a higher-dimensional space, where the resulting matrix exhibits low-rank properties, and kernel-based techniques are applied to achieve data recovery. In [18], a network embedding-based method is introduced for recovering missing measurements in power systems, where spatial and temporal graphs are constructed to represent spatial and temporal dependencies, and the spatiotemporal graph is mapped into low-dimensional spatiotemporal features. Furthermore, in [19], the physics-informed graph modeling with individualized dynamics (Dynamic PIGMind) method is proposed, which achieves high-precision interpretable modeling through physically guided graph structures and node-specific embeddings. In [20], the dynamic causal mechanism learning diagnostics framework (DCMLDF) is proposed, which realizes causal mechanism modeling of dynamic systems through an end-to-end architecture combining causal graph inference and predictive decoding. The above methods provide important references for this paper.

However, existing methods still have certain limitations: (1) Heterogeneous topological representation between branch measurements and bus measurements. Power distribution systems involve data from multiple sources and of multiple types, with different measurement configuration principles and numbers, that is, measurement points do not strictly correspond one-to-one with physical buses. Moreover, traditional graph structures treat branches as edge features, which cannot adapt to the ever-changing nature of branch measurements; (2) Failure to fully utilize the physical coupling relationship between heterogeneous measurements (current/voltage/power). Distribution systems are governed by physical laws such as Kirchhoff's laws and Ohm's law, while purely data-driven methods face limitations including physical infeasibility and insufficient generalization capability. Physics-guided approaches embed physical constraints into the learning process to steer the model toward solutions that are physically feasible and consistent. Therefore, a "physics + data" fusion paradigm is essential; (3) The online application stage failed to effectively integrate real-time available measurement data. Substantial real-time data becomes available during the online operation of power distribution systems, whereas existing approaches inadequately exploit this accessible data, consequently constraining the recovery performance of the model.

To address these challenges, we propose a physics-guided spatio-temporal graph neural network with online data fusion (PSTG-ODF) framework for coordinated recovery of heterogeneous missing measurement data in power distribution systems. The main contributions of this article are as follows:

1) A heterogeneous measurement collaborative recovery mechanism based on multi-channel graph construction and physical constraint coupling is proposed, and a physics-guided spatio-temporal graph neural network-based preliminary recovery model is developed. This model adopts a multi-channel graph construction strategy, building independent weighted graphs for each type of measurement and employing a parallel spatio-temporal graph neural network architecture to independently model the spatio-temporal correlations of measurement data in each channel. On this basis, a physics-guided module is constructed based on physical constraints such as Ohm's law and Kirchhoff's laws, tightly coupling different types of measurements through physical constraints to achieve mutual calibration and joint recovery, thereby significantly improving recovery accuracy.
2) An online data adaptive fusion mechanism based on transparency mask is designed, and a secondary recovery model based on online data fusion is constructed at the model application stage. This model introduces a transparency mask to dynamically evaluate the data credibility of each measurement point, takes the weighted preliminary recovery deviation with the mask as a penalty term, and jointly constructs a secondary optimization model together with the degree of deviation from physical laws, thereby achieving adaptive fusion between preliminary recovery values and available online measurements, further improving recovery accuracy.
3) Experiments on the IEEE 33-node power distribution system and actual power distribution systems demonstrate the physics-guided spatio-temporal graph neural network with online data fusion model's effectiveness in recovering heterogeneous missing measurement data. This paper also analyzes key modules (spatio-temporal prediction, physics guidance, online data fusion) and the effects of missing period length and missing rate. Furthermore, the timeliness of the proposed model is analyzed. Furthermore, the versatility and engineering performance of the proposed model are analyzed.

Before proceeding with the specific research, this section provides a brief explanation of the core terminologies involved in this paper: (1) Physics-guided spatio-temporal graph neural network (PSTG). The core framework of this paper, which embeds physical constraints (such as

Kirchhoff's laws and Ohm's law) into the STGNN architecture in a post-processing optimization manner, enabling collaborative modeling of the spatio-temporal correlations and physical dependencies among multiple types of measurements; (2) Transparency mask. A mask generation mechanism designed in this paper that generates dynamic weight masks by evaluating data credibility to support data fusion; (3) Online data fusion (ODF). The online data fusion mechanism adopted in this paper, designed based on transparency masks and optimization problems, to achieve effective fusion of preliminary recovery values and online available data.

The rest of the paper is organized as follows. Section 2 introduces the preliminary work. Section 3 details the PSTG-ODF model, emphasizing its physics-guided module and data fusion mechanism. Section 4 validates the model via IEEE 33-node system simulations. Finally, Section 5 concludes this paper.

2. Preliminary

From the perspective of measurement device configuration, a relatively complete distribution system measurement configuration is shown in Fig. 1. including line current magnitude (IL), line active power (PL) and reactive power (QL), node voltage magnitude (UN), node injected active power (PN) and reactive power (QN). However, due to investment and maintenance constraints, the completeness of measurement device configuration cannot be fully guaranteed, which further complicates the data recovery process.

Power distribution systems exhibit two primary data loss patterns: (1) sporadic missing from device/communication failures, creating isolated gaps; (2) continuous missing due to communication latency, delaying data delivery to distribution management system (DMS). For instance, advanced metering infrastructure (AMI)-collected consumption/voltage data often experience delays exceeding one hour when reaching DMS, hindering state estimation and operational simulation. Continuous missing cause STGNN error accumulation, severely degrading recovery performance.

3. Proposed methodology

This paper proposes the PSTG-ODF method to jointly model spatio-temporal correlations and physical constraints of heterogeneous measurements while efficiently utilizing online available data. As shown in Fig. 2, the proposed framework has two key stages: model training and model application.

During the model training stage, the proposed PSTG primarily consists of two key modules: a multi-channel spatio-temporal prediction module centered on the STGNN, and a physics-guided post-processing optimization module based on physical laws. First, historical measurements undergo preprocessing and normalization before being categorized and fed into the corresponding channel of the spatio-temporal prediction blocks. The block extracts spatial correlations through GCN and temporal patterns via LSTM to generate predicted values. On this basis, a standalone, lightweight physics-guided post-processing optimization block incorporating Kirchhoff's laws and Ohm's law is subsequently constructed. The preliminary recovery optimization model jointly optimizes measurement predictions to produce preliminary recovered values. Subsequently, by comparing preliminary and actual values, loss functions are computed to update spatio-temporal block parameters in each channel through backpropagation. Through iterative training on the sample set, the final PSTG-based preliminary recovery model is obtained.

During the model application stage, preprocessed historical measurements are fed into the trained PSTG model to generate preliminary recovery values. For available online measurements, transparency mask-based weighted deviations of preliminary recovery values and actual measurements are computed. These deviations are incorporated as penalty terms into the optimization objective, forming a secondary recovery model that fuses preliminary recovery values with online available measurements to produce final recovered values.

3.1. Input block

Within the input block, typical power distribution system measurements (IL, PL, QL, UN, PN and QN) are structured using a sliding-window approach. The input feature matrix for a single time window can be denoted as $X^{(*)} \in R^{N_* \times P}$, containing node measurement values across multiple historical time intervals.

For different types of measurements, this paper adopts a multi-channel graph construction strategy to build independent weighted graph $G^{(*)}=(V^{(*)}, E^{(*)}, A^{(*)})$ for each type of measurement. Essentially, this is a static physical topology graph based on the global positioning system (GPS) and physical connectivity. Here, $V^{(*)}$ denotes the set of measurement points for the corresponding class, $V^{(*)} = \{v_1, \cdots, v_i, \cdots, v_{N_*}\}$, where key equipment points in the power distribution system (such as those measuring current, voltage, etc.) are defined as graph nodes, with their geographic locations determined by GPS coordinates. It is worth noting that measurement points do not strictly correspond one-to-one with the physical nodes (buses) of the power distribution system. Different measurements at the same physical node are treated as independent nodes belonging to different graphs. The set of edges $E^{(*)}$ characterizes the connectivity between various measurement points of the corresponding type, and the edges in $E^{(*)}$ can be described as $e = (v_i, v_j) \in E^{(*)}$, meaning that edges are established between measurement points directly connected by physical lines (e.g., overhead lines, cables) according to the single-line diagram of the power distribution system. The weighted adjacency matrix $A^{(*)} \in R^{N_* \times N_*}$ represents the correlation between measurement points, and is calculated as follows:

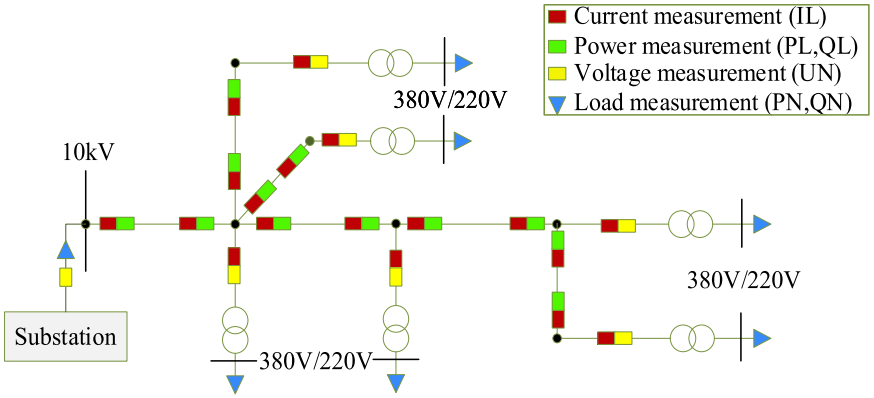


Fig. 1. Typical measurement configurations for medium-voltage power distribution system.

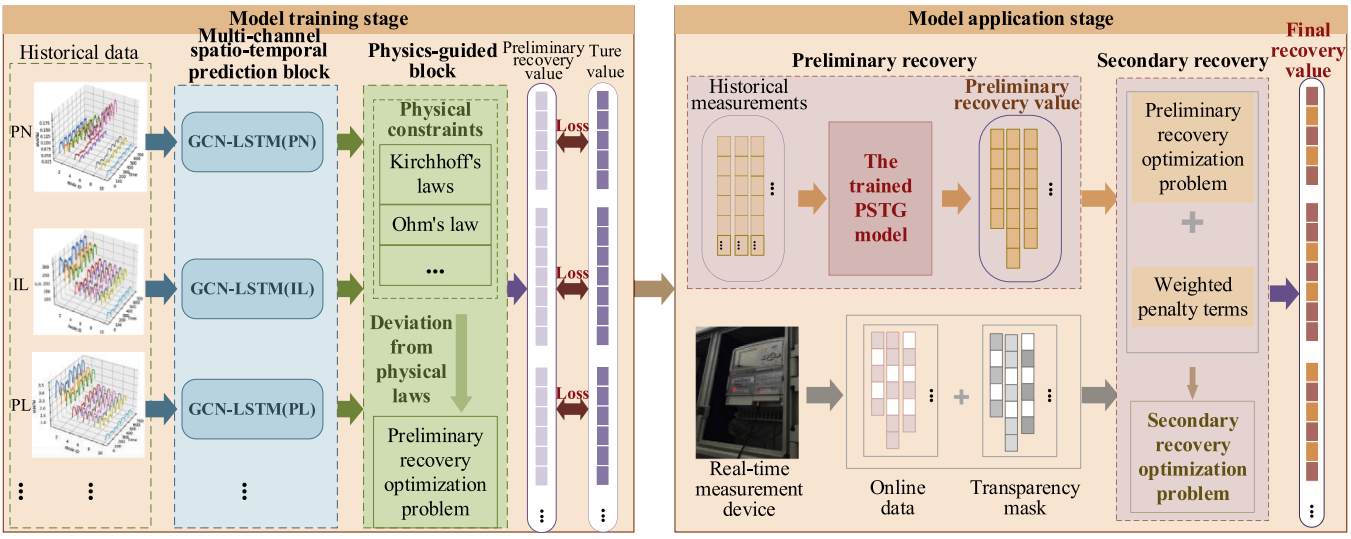


Fig. 2. Proposed PSTG-ODF framework, consisting of a multi-channel spatio-temporal prediction module, a physics-guided post-processing optimization module, and a secondary recovery module.

$$A_{ij}^{(*)} = \begin{cases} 1 & (v_i, v_j) \in E^{(*)} \& (v_i, v_j) \text{isnotabranchcircuit} \\ z_{ij} & (v_i, v_j) \in E^{(*)} \& (v_i, v_j) \text{isabranchcircuit} \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

3.2. PSTG-based preliminary recovery model

1) Multi-Channel Spatio-Temporal Prediction Block

The distribution system data recovery problem studied in this paper inherently forms a graph structure based on its physical entities (measurement points and branches). Furthermore, the features of each measurement point are strongly influenced by its topological neighbors, resulting in significant spatial correlations. Graph neural networks (GNNs), as the preferred framework for processing such non-euclidean structured data, effectively capture spatial correlations among measurement data through a message-passing mechanism that aggregates neighborhood information.

Moreover, measurement data in power distribution systems are collected as time series at fixed sampling intervals, exhibiting pronounced temporal regularity, periodicity, and nonlinear characteristics. as an improved variant of recurrent neural network (RNN), LSTM is particularly well-suited for processing time-series data due to its gating mechanisms and unique memory functionality.

Therefore, this paper combines GNN and LSTM to construct a multi-channel spatio-temporal prediction module, capturing in parallel spatial dependencies determined by graph structures and temporal dependencies determined by measurement data sequences within each type of measurement. Each channel can be considered as learning a complex nonlinear mapping function f on the premise of corresponding graph $G^{(*)}$ and feature matrix $X^{(*)}$, and then predicting the measurement data $\widehat{Y}_T^{(*)} \in R^{N_* \times 1}$ of that category at the current time step, which can be expressed as:

$$\widehat{Y}_T^{(*)} = f(G^{(*)}; X^{(*)}) = f\left(G^{(*)}; \left(X_{T-P}^{(*)}, \cdots, X_{T-1}^{(*)}\right)\right) \tag{2}$$

a) GCN-based spatial dependence modeling

GCN extend the convolutional operation from euclidean to graph-structured data by aggregating features from both the target measurement point and its adjacent nodes to generate refined representations [21].

The graph convolution operator is defined as:

$$H^{(k)} = \mathrm{ELU}\left(L^{\mathrm{sym}} H^{(k-1)} W^{(k)}\right), k = 1, 2, \cdots, K \tag{3}$$

$$\widetilde{A} = A + I \tag{4}$$

$$L^{\mathrm{sym}} = \widetilde{D}^{-\frac{1}{2}} \widetilde{A} \widetilde{D}^{-\frac{1}{2}} \tag{5}$$

where $H^{(k-1)}$ denotes the input of the k-th GCN layer, $H^{(0)} = X^{(*)} \in R^{N_* \times P}$, $\widetilde{D}$ is the diagonal matrix of $\widetilde{A}$.

The spatial block's output is denoted as $H^{(K)} = \{H_1^{(K)}, \cdots, H_l^{(K)}, \cdots, H_L^{(K)}\}$, which is also the input to the temporal module, where L denotes the neuron count in the K-th GCN layer, and $H_l^{(K)} \in R^{N_* \times 1}$ represents the output feature of the l-th neuron.

b) LSTM-based temporal dependence modeling

In this paper, the temporal correlation of the measurement data is modeled using the LSTM model. The components of LSTM include the cell state, forget gate, input gate, and output gate [22]. The basic structure is illustrated in Fig. 3.

The formulas are as follows [23]:

$$f_t = \sigma\left(W_f [h_{t-1}, x_t] + b_f\right) \tag{6}$$

$$i_t = \sigma(W_i [h_{t-1}, x_t] + b_i) \tag{7}$$

$$\widetilde{C}_t = \tanh(W_c [h_{t-1}, x_t] + b_c) \tag{8}$$

$$C_t = f_t \circ C_{t-1} + i_t \circ \widetilde{C}_t \tag{9}$$

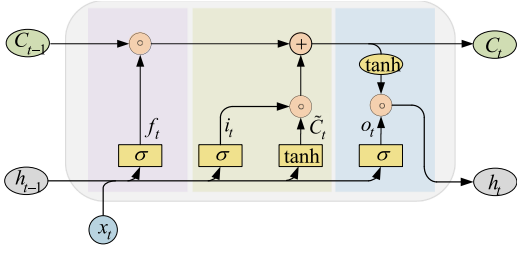


Fig. 3. Structure of LSTM.

$$o_t = \sigma((W_o[h_{t-1}, x_t] + b_o) \tag{10}$$

$$h_t = o_t \circ \tanh(C_t) \tag{11}$$

where $\circ$ denotes the Hadamard product.

c) Fully connected output module

For regression tasks, the LSTM's final hidden state $h_{L'} \in R^{L' \times 1}$ (where L' is the neuron count in the last layer) encodes the complete sequence representation. A fully-connected layer projects this representation to the target space, yielding prediction $\widehat{Y}_T^{(*)} \in R^{N_* \times 1}$:

$$\widehat{Y}_T^{(*)} = W_y h_{L'} + b_y \tag{12}$$

2) Physics-Guided Block

Physics-guided deep learning generally integrates physical knowledge through loss function regularization or neural architecture design via information flow embedding or customized structures [24]. Nevertheless, these approaches encounter problems such as regularization weight determination difficulties, challenges in adaptive optimization control of multiple physical constraints, hyperparameter tuning complexities, and high dependence on physical information accuracy, which restrict their applicability in complex cases.

Therefore, this paper introduces a physics-guided block at the output of the spatio-temporal prediction block. By treating physical constraints as a post-processing optimization layer rather than internal embeddings, the method maintains model lightness while enhancing physical consistency. A schematic illustration is shown in Fig. 4, where $f_{\text{Phy}}(\cdot)$ represents the physical information post-processing procedure. This block utilizes physical constraints (Ohm's and Kirchhoff's laws), with the optimization objective of minimizing deviations from physical laws, to coordinately optimize various measurement predictions $\widehat{Y}_T^{(*)}$ (current, power, voltage) from the spatio-temporal block, yielding preliminary recovery results $\overline{Y}_T^{(*)}$ to improve physical consistency.

This paper achieves physical guidance through a physics-informed post-processing optimization approach, motivated by the following three key considerations: 1) Avoiding the weight tuning problem when physical constraints are used as regularization terms; 2) Maintained the stability of the STGNN training process; 3) Make the physical guidance process more interpretable.

a) Physics deviation metric function

Given the strongly nonlinear constraints imposed by electrical laws on distribution system measurements, we develop physics deviation metrics to assess the violation degree of recovered values against physical laws. In terms of the selection of physical laws, distribution systems involve diverse physical characteristics. However, introducing excessive laws would significantly increase model complexity. Since the proposed data recovery method aims to integrate data-driven and physics-driven approaches, a balance must be struck between recovery performance and computational efficiency.

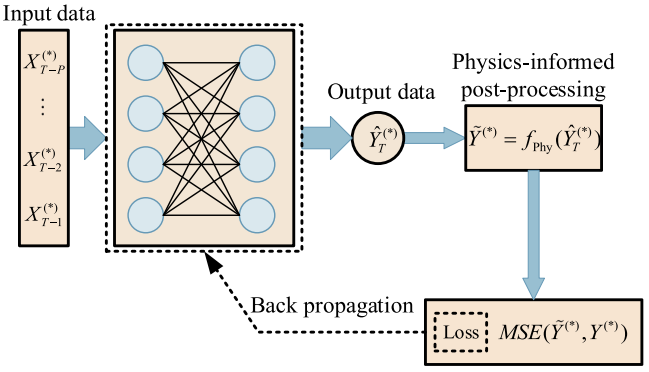


Fig. 4. Physics-guided post-processing optimization model.

Considering that Kirchhoff's laws and Ohm's law are foundational for describing topological connectivity and energy conservation in distribution systems, and that these two laws along with their derived principles have proven effective in establishing intrinsic relationships among various measurements in data recovery scenarios. They are sufficient to appropriately guide the model from a purely data-driven black box toward a physically feasible solution space. Meanwhile, this approach avoids convergence difficulties caused by over-parameterization or excessive nonlinearity in the model. Balancing model performance and computational efficiency, this paper adopts these two laws as the underlying logic and design principle, guiding and structuring the model framework through both direct and indirect forms.

The deviation functions for respective physical laws are defined below.

Considering the relatively low voltage level and short line distances in power distribution systems, the shunt capacitance effect is disregarded. By enforcing Kirchhoff's Current Law on branch (m, j), the deviation function Φ_1 is calculated as:

$$\Phi_1 = \sum_{j \in \Theta_m} \frac{\sqrt{\left(I'_{mj} - I''_{mj}\right)^2}}{\left(I'_{mj} + I''_{mj}\right)/2} \tag{13}$$

By enforcing Kirchhoff's Current Law at node m, the deviation function Φ_2 is calculated as:

$$\Phi_2 = \frac{\sqrt{\left(\sum_{i \in \Psi_m} I''_{im} - I_m - \sum_{j \in \Theta_m} I'_{mj}\right)^2}}{I_m} \tag{14}$$

Disregarding voltage phase angle changes, the active and reactive power deviation functions Φ_3 and Φ_4 for the nodal power injection balance equation at node m are calculated as:

$$\Phi_3 = \frac{\sqrt{\left(P_m - U_m \sum_{j \in m} U_j G_{mj}\right)^2}}{P_m} \tag{15}$$

$$\Phi_4 = \frac{\sqrt{\left(Q_m - U_m \sum_{j \in m} U_j \left(-B_{mj}\right)\right)^2}}{Q_m} \tag{16}$$

where $j \in m$ represents node j connected to node m in the power distribution system, including the self-case ($j = m$).

The voltage U_m can be approximated by:

$$U_m = \frac{\sqrt{P_m^2 + Q_m^2}}{I_m} \tag{17}$$

The active and reactive power deviation functions Φ_5 and Φ_6 for the branch circuit power loss equation of branch (m, j) are calculated as:

$$\Phi_5 = \sum_{j \in \Theta_m} \frac{\sqrt{\left(P'_{mj} - P''_{mj} - I_{mj}^2 R_{mj}\right)^2}}{\left(P'_{mj} + P''_{mj}\right)/2} \tag{18}$$

$$\Phi_6 = \sum_{j\in\Theta_m} \frac{\sqrt{\left(Q'_{mj} - Q''_{mj} - I^2_{mj}X_{mj}\right)^2}}{\left(Q'_{mj} + Q''_{mj}\right)/2} \tag{19}$$

where $I_{mj} = \left(I'_{mj} + I''_{mj}\right)/2$.

The active and reactive power deviation functions Φ_7 and Φ_8 for nodal inflows and outflows are calculated as:

$$\Phi_7 = \frac{\sqrt{\left(\sum_{i\in\Psi_m} P''_{im} - P_m - \sum_{j\in\Theta_m} P'_{mj}\right)^2}}{P_m} \tag{20}$$

$$\Phi_8 = \frac{\sqrt{\left(\sum_{i\in\Psi_m} Q''_{im} - Q_m - \sum_{j\in\Theta_m} Q'_{mj}\right)^2}}{Q_m} \tag{21}$$

The voltage deviation function Φ_9 derived from the apparent power equation is calculated as:

$$\Phi_9 = \frac{\sqrt{\left(U_m - \frac{\sqrt{P_m^2+Q_m^2}}{I_m}\right)^2}}{U_m} \tag{22}$$

b) Preliminary recovery optimization objective

During preliminary recovery stage, the optimization variables consist of all measurement predictions generated by the spatio-temporal prediction block. The optimization objective differs according to the measurement configuration in each recovery scenario, with three representative cases analyzed in this work. As certain optimization models depend on outputs from other models as prior knowledge, we establish two-stage optimization models as follows.

Recovery scenario one: with measurements of node injected active power (PN) and reactive power (QN), and line current magnitude (IL) [25], the two-stage preliminary recovery optimization objectives are constructed as follows:

$$\min J_1 = \sum_{m=1}^{S} (\Phi_1 + \Phi_2) \tag{23}$$

$$\min J_2 = \sum_{m=1}^{S} (\Phi_3 + \Phi_4) \tag{24}$$

Recovery scenario two: with measurements of node injected active power (PN) and reactive power (QN), line current magnitude (IL), and line active power (PL) and reactive power (QL) [26], the two-stage preliminary recovery optimization objectives are constructed as follows:

$$\min J_3 = \sum_{m=1}^{S} (\Phi_1 + \Phi_2 + \Phi_5 + \Phi_6) \tag{25}$$

$$\min J_4 = \sum_{m=1}^{S} (\Phi_7 + \Phi_8) \tag{26}$$

Recovery scenario three: with measurements of node injected active power (PN) and reactive power (QN), line current magnitude (IL), line active power (PL) and reactive power (QL), and node voltage magnitude (UN) [27,28], the two-stage preliminary recovery optimization objectives are constructed as follows:

$$\min J_5 = \min J_3 = \sum_{m=1}^{S} (\Phi_1 + \Phi_2 + \Phi_5 + \Phi_6) \tag{27}$$

$$\min J_6 = \sum_{m=1}^{S} (\Phi_7 + \Phi_8 + \Phi_9) \tag{28}$$

Although each deviation function carries distinct physical meanings, they are all designed in a dimensionless relative error form ((13)-(22)), ensuring preliminary numerical comparability. Meanwhile, we consider that when clear prior preference information is lacking, equal weight setting is a common and reasonable initial choice [29,30]. Simultaneously, to avoid introducing excessive hyperparameters that would overly complicate the model, each deviation function is directly assigned a unified weight in the optimization objective ((23)-(28)).

c) Preliminary recovery constraints

The preliminary recovery optimization is constrained by allowable variation ranges of all decision variables. Taking P_m as an example, the constraints are as follows, and the same applies to others.

$$s.t.\{P_m(1-\lambda) \le P_m \le P_m(1+\lambda) \tag{29}$$

where threshold λ constrains the measurement variation range (empirically set to λ=0.1 in this paper).

d) Solution to the recovery optimization problem

Tailored preliminary recovery optimization models are developed for each distinct scenario. The deviation functions involved in these optimization models are built upon fundamental physical laws of the power system, ensuring rigorous mathematical formulation. However, given that the core objective of the physics-guided module in this paper lies in its orientational role rather than enforceability, it employs physical laws as guides rather than pursuing theoretically absolute optimal solutions. Consequently, we adopt a soft constraint strategy for implementing physical constraints, rather than treating physical laws as hard constraints that must be strictly satisfied.

To balance training efficiency and numerical stability, we introduce necessary simplifications in solving the optimization problem for this module: (1) Constraint simplification: The constraints are set as allowable variation ranges for the optimization variables to limit adjustment amplitudes per iteration and prune the feasible domain ((29)). (2) High-quality initialization: The prediction outputs from the GCN-LSTM are used as initial values, providing the solver with a high-quality starting point to accelerate convergence. (3) Iteration control: The number of optimization iterations is limited to achieve a balance between accuracy and efficiency.

The simplified design described above results in optimization models that are structurally straightforward and fast to solve. Therefore, we employ the classic sequential least squares quadratic programming (SLSQP) algorithm for solving these models. The SLSQP algorithm is particularly well-suited for optimization problems with nonlinear constraints, as it decomposes the original problem into a series of small-scale quadratic programming subproblems, offering advantages in memory usage and computational speed. This algorithm is sufficiently effective for addressing the optimization tasks in this study, and its characteristics align well with the module's requirements for efficiency and simplicity.

3) Error Back Propagation

Regarding the preliminary recovery values $\overline{Y}_T^{(*)}$ (current, power, voltage) generated by the physics-guided block, the following loss functions are formulated separately:

$$Loss^{(*)} = MSE(\widetilde{Y}^{(*)}, Y^{(*)}) \tag{30}$$

where $Loss^{(*)}$ denotes the loss function for measurement *, $MSE(\widetilde{Y}^{(*)}$,

$Y^{(*)}$) represents the mean squared error between the preliminary recovery values and true values of measurement *.

The gradients of all measurement-specific loss functions are back-propagated through the network via the chain derivation rule, subsequently updating the spatio-temporal prediction block's parameters (weights and biases).

3.3. ODF-based secondary recovery model

To effectively utilize online available measurement data for enhanced repair accuracy, this paper proposes an ODF-based secondary recovery model. First, the credibility of available measurements is evaluated using Dempster-Shafer (D-S) evidence theory, based on which transparency masks are established. Then, weighted penalty terms of preliminary recovery values and online measurements are constructed based on transparency masks, and integrated with preliminary recovery objectives to form the secondary recovery optimization model. Finally, the secondary optimization model is solved to obtain final recovered values $\widetilde{Y}_T^{(*)} \in R^{N_s \times 1}$. The schematic of the secondary recovery is shown in Fig. 5.

1) Calculation of Transparency Mask

Let $\xi_T^{(*)} = \{\xi_T^i\} \in R^{N_s \times 1}$ denote the online measurement data with missing values, where a 0–1 mask identifies their missing positions. Meanwhile, accounting for measurement device errors, communication noise, and environmental interference in practical systems, measurement credibility is typically below 1 (100%). Therefore, we introduce transparency $\vartheta \in [0,1]$ to quantify measurement credibility, and construct transparency mask $M_T^{(*)} = \{m_T^i\} \in R^{N \times 1}$ defined as:

$$m_T^i = \begin{cases} \vartheta & \text{if} \xi_T^i \text{isobserved} \\ 0 & \text{if} \xi_T^i \text{ismissing} \end{cases} \tag{31}$$

Based on the dempster-shafer (D-S) evidence theory [31,32], this paper assigns a transparency value between [0,1] to each available measurement point by fusing multiple sources of evidence, serving as the basis for subsequent weighted fusion. The basic process includes identification framework determination, basic belief assignment function, Dempster credibility synthesis, and transparency value calculation.

a) Identification framework determination

This paper selects six credibility assessment indicators: measurement device error ζ_1, device failure rate ζ_2, device aging level ζ_3, environmental factor ζ_4, historical data missing ratio ζ_5 and historical data validity ζ_6. Their mathematical formulations are detailed in Appendix A. According to evaluation rules, measurement credibility is categorized into: credible $\Re_1$, moderately credible $\Re_2$, marginally credible $\Re_3$, not credible $\Re_4$, as shown in Table 1.

b) Basic belief assignment function

The basic belief assignment function $\mathbb{F}_\alpha(\cdot)$ quantifies the support degree of credibility evaluation metrics to each credibility level under uncertainty, consisting of a deterministic component $\mathbb{F}_\alpha(\Re_\beta)$ and an uncertain component $\mathbb{F}_\alpha(\Re_0)$. The calculation formulas are given in Appendix B.

c) Credibility synthesis

Dempster's combination rule is applied to fuse basic belief assignment function $\mathbb{F}_\alpha(\cdot)$ through orthogonal sum operations, obtaining the final belief degree $\mathbb{F}(\Re_\beta)$ for all credibility levels. $\beta = 1, 2, 3, 4$. The calculation formula is given in Appendix B.

From the belief degrees corresponding to each credibility level mentioned above, the transparency value ϑ can be calculated as follows:

$$\mathbb{F}_{\max} = \max(\mathbb{F}(\Re_1), \mathbb{F}(\Re_2), \mathbb{F}(\Re_3), \mathbb{F}(\Re_4)) \tag{32}$$

$$\vartheta = \begin{cases} 0.9 & \text{if} \mathbb{F}(\Re_1) = \mathbb{F}_{\max} \\ 0.7 & \text{if} \mathbb{F}(\Re_2) = \mathbb{F}_{\max} \\ 0.5 & \text{if} \mathbb{F}(\Re_3) = \mathbb{F}_{\max} \\ 0 & \text{if} \mathbb{F}(\Re_4) = \mathbb{F}_{\max} \end{cases} \tag{33}$$

This mapping follows the principle of conservative estimation. When there is sufficient evidence supporting credibility, the transparency

Table 1
Measurement credibility levels.

Evaluation indicators	Credibility level			
	$\Re_1$	$\Re_2$	$\Re_3$	$\Re_4$
ζ_1	[0, 2%)	(1%, 6%)	(4%, 10%)	[10%, 1]
ζ_2	[0, 15%)	(10%, 25%)	(20%, 40%)	[40%, 1]
ζ_3	[0, 2%)	(1%, 4%)	(3%, 6%)	[6%, 1]
ζ_4	(0.8, 1]	(0.6, 0.9)	(0.4, 0.7)	[0, 0.4)
ζ_5	[0, 15%)	(10%, 25%)	(20%, 40%)	[40%, 1]
ζ_6	[0, 15%)	(10%, 25%)	(20%, 40%)	[40%, 1]

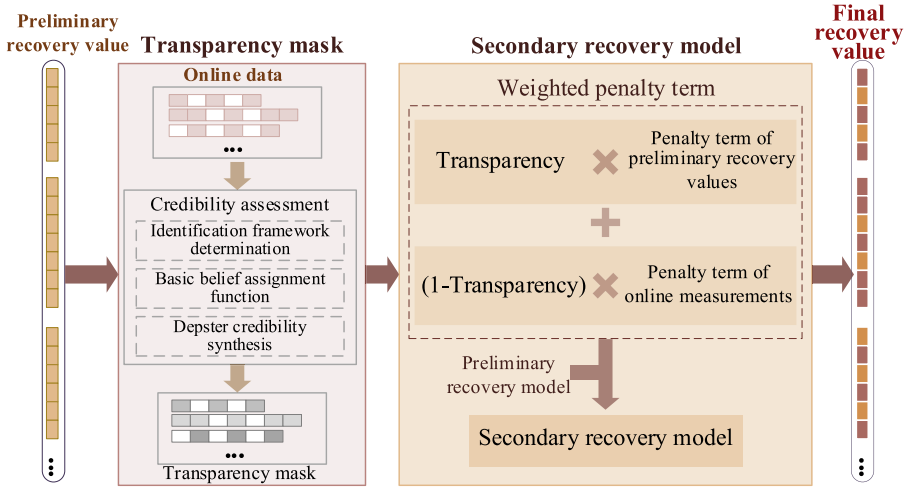


Fig. 5. Schematic of secondary recovery.

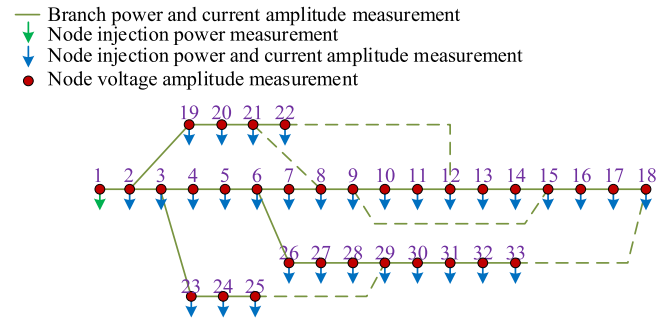


Fig. 6. IEEE 33-node power distribution system.

value is set to 0.9; when evidence is uncertain or conflicting, transparency is appropriately lowered (to 0.7 or 0.5); and when evidence consistently points to a lack of credibility, transparency is set to 0.

2) Secondary Recovery Optimization Problem

The preliminary fusion of PSTG's preliminary recovery values $\overline{Y}_T^{(*)}$ and online measurements $\xi_T^{(*)}$ is formulated as:

$$\theta_0^{(*)} = M_T^{(*)} \odot \xi_T^{(*)} + \left(1 - M_T^{(*)}\right) \odot \overline{Y}_T^{(*)} \tag{34}$$

where the symbol $\odot$ denotes element-wise multiplication.

The initial fused value $\theta_0^{(*)}$ constitutes the original solution vector for the secondary recovery optimization problem, and the secondary optimization solution vector for the k-th iteration is denoted as $\theta_k^{(*)} \in R^{N_* \times 1}$. The weighted penalty term for the k-th iteration is then constructed:

$$\Gamma_k^{(*)} = \Gamma_{k,1}^{(*)} + \Gamma_{k,2}^{(*)} = \vartheta^{(*)} \parallel \frac{M_T^{(*)}}{\vartheta^{(*)}} \odot \left(\xi_T^{(*)} - \theta_{k-1}^{(*)}\right) \parallel + (1 - \vartheta^{(*)}) \parallel \left(1 - \frac{M_T^{(*)}}{\vartheta^{(*)}}\right) \odot \left(\overline{Y}_T^{(*)} - \theta_{k-1}^{(*)}\right) \parallel \tag{35}$$

where $\| \cdot \|$ denotes the Euclidean paradigm, $\Gamma_{k,1}^{(*)}$ and $\Gamma_{k,2}^{(*)}$ quantify the deviations of online measurements and preliminary recovery values, respectively.

Based on this, according to the classification of optimization variables, the formulated weighted penalty function is incorporated to modify the preliminary recovery objectives ((23)-(28)) during the training stage, yielding the secondary recovery objectives ((36)-(41)). The variable definitions and constraint formulations remain consistent with the preliminary recovery optimization model.

$$\min J_1' = \sum_{m=1}^{S} (\Phi_1 + \Phi_2) + \Gamma_k^{(\mathrm{IL})} \tag{36}$$

$$\min J_2' = \sum_{m=1}^{S} (\Phi_3 + \Phi_4) + \Gamma_k^{(\mathrm{PB})} + \Gamma_k^{(\mathrm{QB})} \tag{37}$$

$$\min J_3' = \sum_{m=1}^{S} (\Phi_1 + \Phi_2 + \Phi_5 + \Phi_6) + \Gamma_k^{(\mathrm{IL})} + \Gamma_k^{(\mathrm{PL})} + \Gamma_k^{(\mathrm{QL})} \tag{38}$$

$$\min J_4' = \sum_{m=1}^{S} (\Phi_7 + \Phi_8) + \Gamma_k^{(\mathrm{PB})} + \Gamma_k^{(\mathrm{QB})} \tag{39}$$

$$\min J_5' = \min J_3' \tag{40}$$

$$\min J_6' = \sum_{m=1}^{S} (\Phi_7 + \Phi_8 + \Phi_9) + \Gamma_k^{(\mathrm{PB})} + \Gamma_k^{(\mathrm{QB})} + \Gamma_k^{(\mathrm{UB})} \tag{41}$$

The convergence analysis of the preliminary recovery optimization and secondary recovery optimization problems is detailed in Appendix C.

3.4. Evaluation metrics

This paper evaluates the model recovery accuracy by several evaluation metrics: root mean square error (RMSE), mean absolute error (MAE), and mean absolute percentage error (MAPE), specifically defined as follows.

$$RMSE = \sqrt{\frac{1}{N_* \mathrm{T}_*} \sum_{i=1}^{N_*} \sum_{t=1}^{\mathrm{T}_*} \left(\widetilde{Y}_{it} - Y_{it}\right)^2} \tag{42}$$

$$MAE = \frac{1}{N_* \mathrm{T}_*} \sum_{i=1}^{N_*} \sum_{t=1}^{\mathrm{T}_*} \left|\widetilde{Y}_{it} - Y_{it}\right| \tag{43}$$

$$MAPE = \frac{100\%}{N_* \mathrm{T}_*} \sum_{i=1}^{N_*} \sum_{t=1}^{\mathrm{T}_*} \left|\frac{\widetilde{Y}_{it} - Y_{it}}{Y_{it}}\right| \tag{44}$$

Table 2
Comparison of recovery performance of different models under various scenarios ($\tau = 1, \eta = 25\%$).

Measurement type	Metric	DCRNN	AGCRN	STGCN	STSGCN	BSVT	Softwalk	SAITS	CSDI	PSTG-ODF		
										Scenario one	Scenario two	Scenario three
IL	RMSE (A)	1.1715	1.0426	1.1300	1.1068	0.9305	0.9485	0.8475	0.7932	0.5900	0.5775	**0.5757**
	MAE (A)	0.7138	0.6357	0.6887	0.6746	0.5638	0.5781	0.5142	0.4813	0.3614	0.3538	**0.3527**
	MAPE (%)	2.8444	2.5317	2.7439	2.6875	2.2637	2.3092	2.0873	1.9565	1.4331	1.4028	**1.3985**
PL	RMSE (MW)	0.0086	0.0080	0.0088	0.0080	0.0067	0.0071	0.0074	0.0069	-	**0.0059**	**0.0059**
	MAE (MW)	0.0056	0.0052	0.0058	0.0052	0.0040	0.0046	0.0048	0.0045	-	**0.0038**	**0.0038**
	MAPE (%)	1.6791	1.5746	1.7239	1.5727	1.4463	1.4834	1.5486	1.4632	-	**1.1752**	1.1806
QL	RMSE (Mvar)	0.0058	0.0053	0.0055	0.0053	0.0049	0.0050	0.0052	0.0048	-	**0.0040**	**0.0040**
	MAE (Mvar)	0.0038	0.0034	0.0036	0.0035	0.0031	0.0033	0.0034	0.0032	-	**0.0026**	**0.0026**
	MAPE (%)	1.7653	1.6091	1.6813	1.6248	1.5209	1.5710	1.6245	1.5381	-	1.2481	**1.2364**
PN	RMSE (MW)	0.0070	0.0062	0.0070	0.0062	0.0055	0.0058	0.0060	0.0054	0.0037	0.0037	**0.0035**
	MAE (MW)	0.0033	0.0029	0.0033	0.0029	0.0024	0.0026	0.0027	0.0023	0.0017	0.0017	**0.0016**
	MAPE (%)	3.3466	2.9794	3.3620	2.9991	2.4526	2.5033	2.5841	2.4217	1.8428	1.8276	**1.7375**
QN	RMSE (Mvar)	0.0098	0.0084	0.0095	0.0090	0.0069	0.0073	0.0074	0.0069	0.0057	0.0058	**0.0056**
	MAE (Mvar)	0.0034	0.0029	0.0033	0.0031	0.0025	0.0027	0.0028	0.0024	0.0020	0.0020	**0.0019**
	MAPE (%)	3.5337	2.9550	3.4030	3.1886	2.4981	2.5381	2.6198	2.4673	1.7556	1.8012	**1.7177**
UN	RMSE (kV)	0.0058	0.0053	0.0051	0.0055	0.0049	0.0051	0.0052	0.0048	-	-	**0.0047**
	RMSE (kV)	0.0037	0.0033	0.0032	0.0034	0.0031	0.0032	0.0032	0.0031	-	-	**0.0029**
	MAPE (%)	0.0314	0.0287	0.0277	0.0297	0.0268	0.0271	0.0271	0.0264	-	-	**0.0255**

where $\tilde{Y}_{it}$ and Y_{it} denote the recovered and true value of measurement point v_i at the t-th missing time step, respectively.

3.5. Implement procedure of PSTG-ODF

The proposed PSTG-ODF model implementation comprises two key phases:

1) Training and validation. The training dataset is processed sequentially through the input block, the spatio-temporal prediction block, and the physics-guided block to generate the preliminary recovery data. Then, the loss function is computed between the preliminary recovery data and truth data, followed by error backpropagation to update the PSTG model parameters. The best-performing model is saved for deployment.
2) Testing. The test dataset is fed into the trained preliminary recovery model to obtain preliminary results. An ODF-based secondary recovery model is then constructed to produce final recovery results. The complete pseudocode of PSTG-ODF is presented in Algorithm 1.

Algorithm 1 Proposed PSTG-ODF Method

Input: training set D, valid set D_{valid}, testing set D_{test} (D=$D^{(\text{IL})}$ or $D^{(\text{PL})}$ or $D^{(\text{QL})}$ or $D^{(\text{PN})}$ or $D^{(\text{QN})}$ or $D^{(\text{UN})}$)
Output: recovery results $\tilde{Y}_{\text{test}}$ of missing measurement data ($\tilde{Y}$=$\tilde{Y}^{(\text{IL})}$ or $\tilde{Y}^{(\text{PL})}$ or $\tilde{Y}^{(\text{QL})}$ or $\tilde{Y}^{(\text{PN})}$ or $\tilde{Y}^{(\text{QN})}$ or $\tilde{Y}^{(\text{UN})}$)
In the training and validation stage:
1. Input Block
Input: training set D, impedance amplitude z
Output: feature matrix X, adjacency matrix A
① Construct feature matrix X for each type of measurement using the sliding window method
② Calculate adjacency matrix A for each type of measurement
for each training epoch **do**
2. Spatio-Temporal Prediction Block
Input: dataset X, adjacency matrix A
Output: GCN-LSTM prediction results $\hat{Y}$
① Calculate graph convolutions, obtain $H^{(K)}$ (output of the last GCN layer)
② Process using LSTM cells, obtain $h_{L'}$ (hidden state of the last time step)
③ Process using full connection layer, obtain prediction results $\hat{Y}$
3. Physics-Guided Block
Input: prediction results $\hat{Y}$
Output: primary recovery results $\overline{Y}$
① Construct optimization objectives and constraints based on measurement configurations
② Solve optimization problems using SLSQP algorithm, obtain primary recovery results $\overline{Y}$
4. Calculate the loss function, and update the entire model with back propagation
5. Calculate the performance indicators of the model on the valid set, and keep the best-trained model
end for
In the testing stage:
1. Primary Recovery
Input: testing set D_{test}
Output: primary recovery results $\overline{Y}_{\text{test}}$
Process using the trained model
2. Secondary Recovery
Input: primary recovery results $\overline{Y}_{\text{test}}$
Output: final recovery results $\tilde{Y}_{\text{test}}$ of missing measurement data
① Construct transparency mask based on online measurement data and its credibility
② Construct weighted penalties for online measurement data and primary recovery data
③ Design and solve secondary recovery optimization problems

4. Case study

The PSTG-ODF method is implemented in Python 3.8 using PyCharm 2023.1 as the integrated development environment. This section presents the numerical simulation results and comparisons conducted on a computer with Intel(R) Core(TM) i7-14700 CPU @ 2.10 GHz and 32GB main memory.

4.1. Test system

This paper uses a standard IEEE 33-node power distribution system as the test system [33], and its structure is shown in Fig. 6. There are a total of 37 branches, including five interconnection switch branches. The measure dataset contains six types of measurements (IL, PL, QL, PN, QN, and UN), where IL has 96 measurement points, PL and QL each have 64 measurement points, and PN, QN, and UN each consist of 33 measurement points (i.e., the number of measurement points $N_{\text{IL}} = 96$, $N_{\text{PL}} = N_{\text{QL}} = 64$, $N_{\text{PN}} = N_{\text{QN}} = N_{\text{PN}} = 33$, and the total number of nodes $S = 33$).

This paper selects the electrical demand data from 32 European countries between June 1, 2018, and August 30, 2018, as the load data [34], with a data interval of 15 min, totaling 8736 sets of time-steps data. Power flow simulations are conducted in OpenDSS using these load profiles, with zero-mean Gaussian noise (the standard deviation is 0.01) added to the power flow solutions to generate the measurement dataset [35]. Then the normalized dataset is divided into training, validation and test sets in a 6:2:2 ratio.

4.2. Experimental setup

1) Experimental Settings

The number of input features is set to $P = 12$, and the adjacency matrix dimensions are determined based on the cardinality of different measurement types: $A^{(\text{IL})} \in R^{96\times96}$ for IL; $A^{(\text{PL})} = A^{(\text{QL})} \in R^{64\times64}$ for PL and QL; $A^{(\text{PN})} = A^{(\text{QN})} = A^{(\text{UN})} \in R^{33\times33}$ for PN, QN and UN. In the physics-guided block, the threshold for constraining measurement value fluctuations is set to λ=0.1. The model hyperparameters are configured as follows: both GCN and LSTM have 2 layers with neuron counts set to (128, 128) and (64, 64) respectively, Adam optimizer, MSE loss function, 500 training epochs, 64 batch size, initial learning rate of 0.001, the maximum optimization iterations for SLSQP is set to 40. In terms of the missing data generation protocol, the missing types are divided into sporadic missing (randomly selecting a portion of measurement points) and continuous missing (randomly selecting consecutive time windows). The random seed is set to 42, and the generation of missing positions is sampled based on a preset missing ratio (e.g., 25%) using functions such as “np.random.choice” and “np.random.randint”.

2) Baseline Methods

We compare PSTG-ODF with the following eight baseline methods, which are currently advanced spatio-temporal graph neural networks, data recovery methods for power systems and time series imputation methods, including:

a) DCRNN [36]: diffusion convolutional recurrent neural network, which integrates diffusion graph convolution with gated recurrent units into an encoder-decoder architecture.
b) AGCRN [37]: adaptive graph convolutional recurrent network, which learns data-adaptive graph structures to model spatial correlations and employs gated recurrent units to capture temporal dependencies.
c) STGCN [38]: spatio-temporal graph convolutional network, which combines graph convolution and gated temporal convolution to capture spatio-temporal dependencies.
d) STSGCN [39]: spatio-temporal synchronous graph convolutional network, which considers spatio-temporal dependencies in local adjacent time steps.
e) BSVT [7]: bayesian singular value thresholding, which incorporates prior knowledge in the form of second-order statistics.
f) Softwalk [18]: a network embedding method for learning low-dimensional feature representations of power grids, employing a softwalking strategy to capture spatio-temporal correlations.

g) SAITS [40]: a self-attention-based imputation model for time series that performs missing value imputation based on Transformer architecture and a masking mechanism.
h) CSDI [41]: a conditional score-based diffusion model for time series imputation that performs probabilistic imputation of missing values in time series by leveraging denoising score matching and conditional generation mechanisms.

4.3. Experimental results

1) Overall Comparison

For three typical recovery scenarios, Table 2 shows the recovery performance comparison between the proposed PSTG-ODF method and eight baseline methods under missing period length $\tau = 1$ (sporadic missing) and missing rate $\eta = 25\%$.

As shown in Table 2:

a) The proposed PSTG-ODF demonstrates statistically superior performance (RMSE/MAE/MAPE) over all baseline methods across diverse distribution system measurements, conclusively validating its effectiveness and advanced recovery capabilities for distribution system data recovery.
b) Compared with spatio-temporal graph neural networks such as DCRNN, AGCRN, STGCN, and STSGCN, the PSTG-ODF model shows particularly significant performance improvement on both PN and QN datasets. For the PN dataset, the proposed method reduces RMSE, MAE and MAPE by 40.3–50.0%, 41.4–51.5% and 38.1−48.3% respectively, demonstrating substantial error reduction. This significant improvement stems from the weak spatial correlations in nodal injected power (primarily driven by consumer behavior), which constrains the effectiveness of baseline methods. In contrast, PN/QN has stronger physical coupling with node voltage, line power, etc., so the contribution of the physical guidance module is more prominent.

2) Effects of Missing Period Length

To systematically evaluate missing period length effects on recovery performance, we conduct controlled experiments on the PN dataset with three distinct missing patterns: $\tau = 1$ (sporadic missing), $\tau = 4$ (continuous missing), and $\tau = 12$ (continuous missing) at missing ratio $\eta = 25\%$. Table 3 presents the comprehensive performance comparison between the PSTG-ODF model and baseline methods in recovery scenario one.

Table 3 demonstrates that the proposed PSTG-ODF achieves optimal RMSE/MAE/MAPE values across all missing period lengths, with its performance advantage over baselines becoming more pronounced as the missing duration increases. For MAPE specifically, the performance margins between PSTG-ODF and baselines grow from 0.5789∼1.5192 percentage points ($\tau = 1$), 1.2062∼3.0183 percentage points ($\tau = 4$) to 1.8146∼4.8389 percentage points ($\tau = 12$). These results confirm PSTG-ODF's ability to maintain superior recovery accuracy during prolonged data outages, demonstrating exceptional algorithmic stability and robustness.

3) Effects of Missing Rate

To investigate the impact of missing rates on model recovery performance, this paper takes the PN dataset in recovery scenario one as an example, setting four missing rates of 10%, 25%, 50% and 75%, and compares the recovery performance of the PSTG-ODF model with several baseline methods under missing period length $\tau = 1$. The results are shown in Fig. 7.

As shown in Fig. 7:

a) Under various missing rates, the PSTG-ODF model achieves the highest recovery accuracy (manifested as the lowest MAPE values). However, at low missing rates (e.g., 10% and 25%), the performance gap between PSTG-ODF and baseline methods is relatively small. This is mainly attributed to the higher data integrity at low missing rates, allowing baseline methods to leverage substantial valid neighboring node data and historical data to achieve relatively accurate recovery.
b) With increasing missing ratios, baseline methods exhibit deteriorating performance (sharply rising MAPE) due to their limited capacity in capturing global data features and complex relationships. In contrast, the proposed PSTG-ODF model benefits from unique physical correlation modeling and online data fusion mechanism. It effectively leverages auxiliary information such as heterogeneous measurement data and partially reliable online data, in addition to fully utilizing extensive historical measurement data. As a result, the model maintains high recovery accuracy even at high missing rates. This is primarily because, when there is a severe lack of spatiotemporal information, the coupling relationship between physical constraints and heterogeneous measurements becomes a key source of supplementary information, reflecting the unique value of "physical guidance" in extreme scenarios.

4) Ablation Analysis

a) Module ablation

To investigate the effectiveness of the key modules in the PSTG-ODF model proposed in this paper, we compare the PSTG-ODF model with the deep neural network (DNN), GCN, LSTM, GCN-LSTM models, and the PSTG model without secondary recovery, with recovery scenario two as an example. Table 4 shows the recovery performance of each model under missing period length $\tau = 1$ and missing rate $\eta = 25\%$.

As shown in Table 4:

a) Compared with the DNN model, all types of recovery errors for the GCN and LSTM models are significantly reduced, validating that appropriate modeling of spatial and temporal correlations can effectively enhance the accuracy of measurement data recovery in power distribution systems.
b) The GCN-LSTM model hybrid demonstrates consistent superiority over standalone GCN (spatial-only) and LSTM (temporal-only) models across most scenarios. For PL dataset, it achieves MAPE reductions of 2.6 percentage points and 0.6 percentage points respectively, confirming the critical role of spatio-temporal joint modeling in measurement data recovery.
c) Even without secondary optimization, PSTG surpasses GCN-LSTM comprehensively. For PN data, it shows improvements of 0.0060MW (RMSE), 0.0029MW (MAE) and 2.3238 percentage points (MAPE), validating the efficacy of its physics-guided module.
d) Compared to the GCN-LSTM model, the proposed PSTG-ODF achieves significant performance gains across all measurement types. For the PN dataset, it reduces RMSE by 0.0066MW, MAE by 0.0034MW, and MAPE by 2.8619 percentage points. Moreover, PSTG-ODF outperforms the PSTG model, further verifying that the PSTG-based preliminary recovery model can greatly improve the recovery performance of power distribution system measurements, and the ODF-based secondary recovery model can further improve recovery accuracy. It should be noted that, as a secondary recovery module, the marginal benefit of the ODF module is affected by the quality of online data. Therefore, in practical deployment, the ODF module holds greater application value in scenarios with high measurement coverage or a high proportion of measurements with low missing rates.

b) Parameter sensitivity.

Table 3
Recovery performance of different models under various missing period lengths on PN datasets in recovery scenario one ($\eta = 25\%$).

Missing period length	Metric	DCRNN	AGCRN	STGCN	STSGCN	BSVT	Softwalk	SAITS	CSDI	PSTG-ODF
$\tau = 1$	RMSE (MW)	0.0070	0.0062	0.0070	0.0062	0.0055	0.0058	0.0060	0.0054	**0.0037**
	MAE (MW)	0.0033	0.0029	0.0033	0.0029	0.0024	0.0026	0.0027	0.0023	**0.0017**
	MAPE (%)	3.3466	2.9794	3.3620	2.9991	2.4526	2.5033	2.5841	2.4217	**1.8428**
$\tau = 4$	RMSE (MW)	0.0329	0.0326	0.0331	0.0326	0.0207	0.0264	0.0214	0.0195	**0.0070**
	MAE (MW)	0.0096	0.0094	0.0096	0.0094	0.0069	0.0073	0.0069	0.0063	**0.0033**
	MAPE (%)	6.3047	6.1241	6.3755	6.1416	4.7365	5.0694	4.8426	4.5634	**3.3572**
$\tau = 12$	RMSE (MW)	0.0421	0.0382	0.0404	0.0399	0.0350	0.0358	0.0369	0.0349	**0.0315**
	RMSE (MW)	0.0141	0.0122	0.0132	0.0130	0.0107	0.0110	0.0113	0.0108	**0.0088**
	MAPE (%)	10.4888	8.7067	9.6871	9.4840	7.4645	7.6256	7.8965	7.5288	**5.6499**

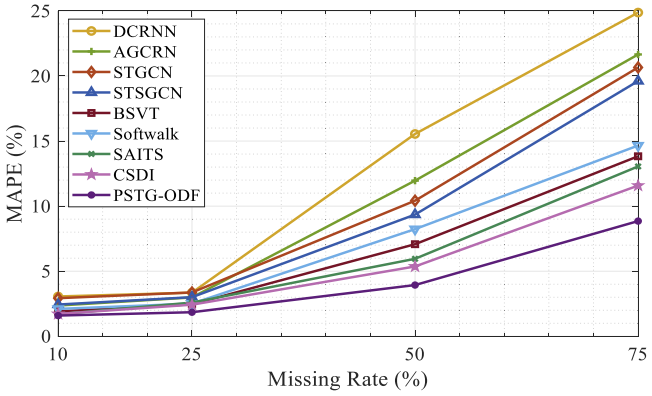


Fig. 7. Recovery MAPE (%) of different methods under different missing rates for PN dataset in recovery scenario one.

We conduct sensitivity experiments on some parameters involved in the physics-guided module (the threshold λ that limits the variation range of measured values, and the maximum number of SLSQP optimizations). Taking the recovery scenario two as an example, we use MAPE value and model training time as evaluation metrics. The performance of the PSTG-ODF model under different threshold λ is shown in Table 5, while the model performance and training time under different maximum SLSQP optimization iterations (Maxiter) are presented in Fig. 8 and Table 6, respectively.

As shown in Table 5, when λ is too small (0.05), the constraints are too tight, limiting the correction capability of the physics-guided optimization, resulting in a relatively large MAPE value. As λ increases, the adjustment range of the optimization variables expands, and the degree of freedom in the physics-guided optimization increases. When λ is in the range of 0.10–0.15, the MAPE value is relatively small. As λ continues to increase, overly loose constraints weaken the role of data-driven methods, leading to a certain increase in MAPE.

As can be seen from Fig. 8 and Table 6:

a) As Maxiter increases from 20 to 80, the MAPE shows an overall downward trend. The most significant decrease in MAPE occurs when Maxiter increases from 20 to 40; when Maxiter increases from 40 to 60, the MAPE continues to decline but at a slower rate; when Maxiter increases from 60 to 80, the MAPE for IL, PL, and QL shows a slight increase, while that for PN and QN continues to decrease modestly.
b) As Maxiter increases, the model training time also increases. In the three stages from Maxiter 20 to 80, the training time increases by 35.47 min, 57.12 min, and 44.66 min, respectively.
c) Overall, the maximum number of SLSQP optimizations has a relatively small impact on the performance of the PSTG-ODF model, but has a significant impact on model training time. In this paper, Maxiter = 40 is selected as the default setting to strike a balance between model performance and computational cost.

In addition, we conducted sensitivity experiments on the credibility levels in the transparency mask computation section, and the experimental results are presented in Appendix D.

5) Versatility and generalization

To investigate the versatility and generalization of the proposed PSTG-ODF model, taking recovery scenario one as an example, the following extensibility validation experiments are designed.

Two 10 kV power distribution systems (XI46 feeder and WAN48 feeder) in a certain area of Tianjin are introduced. The measurement configurations of the XI46 feeder and WAN48 feeder are shown in Figs. 9 and 10, respectively. The sample dataset consists of branch current, node active power and reactive power measurements from March 1, 2022 to May 31, 2022, with a 15-minute data interval, totaling 8832 sets of time-steps data.

Since the actual measurement data itself contains missing values,

Table 4
Effects of major blocks in recovery scenario two ($\tau = 1$,$\eta = 25\%$).

Restoration quantity	Metric	DNN	GCN	LSTM	GCN-LSTM	PSTG	PSTG-ODF
IL	RMSE (A)	6.5385	5.751	1.9974	1.4046	0.7419	**0.5775**
	MAE (A)	2.8901	2.5424	1.2136	0.8513	0.4245	**0.3538**
	MAPE (%)	6.0475	5.3173	4.1521	3.4090	1.6354	**1.4028**
PL	RMSE (MW)	0.0822	0.0697	0.0178	0.0101	0.0063	**0.0059**
	MAE (MW)	0.0409	0.0335	0.0120	0.0068	0.0044	**0.0038**
	MAPE (%)	5.4505	4.4618	2.4434	1.8648	1.3926	**1.1752**
QL	RMSE (Mvar)	0.0511	0.0482	0.0109	0.0068	0.0046	**0.0040**
	MAE (Mvar)	0.0245	0.0226	0.0073	0.0044	0.0029	**0.0026**
	MAPE (%)	5.4022	4.7568	2.6587	1.9974	1.4435	**1.2481**
PN	RMSE (MW)	0.0322	0.0250	0.0103	0.0103	0.0043	**0.0037**
	MAE (MW)	0.0092	0.0069	0.0051	0.0051	0.0022	**0.0017**
	MAPE (%)	5.9919	4.7069	4.9563	4.6895	2.3657	**1.8276**
QN	RMSE (Mvar)	0.0254	0.0122	0.0189	0.0112	0.0068	**0.0058**
	MAE (Mvar)	0.0065	0.0042	0.0047	0.0040	0.0024	**0.0020**
	MAPE (%)	6.7659	4.8714	5.2409	4.8406	2.2481	**1.8012**

Table 5
Recovery performance of PSTG-ODF model under different threshold λ in recovery scenario two ($\tau = 1, \eta = 25\%$).

Restoration quantity	Metric	$\lambda = 0.05$	$\lambda = 0.1$	$\lambda = 0.15$	$\lambda = 0.2$
IL	MAPE (%)	1.4783	1.4028	1.4259	1.4604
PL		1.2367	1.1752	1.1969	1.2276
QL		1.3124	1.2481	1.2690	1.2976
PN		1.9158	1.8276	1.8403	1.8593
QN		1.8872	1.8012	1.8089	1.8262

and there are no true values at the missing positions that can be directly used for error calculation, this paper adopts the following experimental setup: For the training data, cubic spline interpolation is used for preliminary imputation before the data is used for model training. For the test data, first, the proposed method is used to impute the actual missing values in the original data to construct a complete benchmark dataset; then, artificial missing values are introduced into the normal data of this benchmark dataset; finally, the imputation results of each method are compared with the corresponding positions in the benchmark dataset, and error metrics are calculated to evaluate the imputation performance.

For the XI46 feeder and WAN48 feeder, the comparison of imputation performance between the proposed PSTG-ODF method and several baseline methods under scenario one, missing period length $\tau = 1$ and missing rate $\eta = 25\%$ is shown in Table 7 and Fig. 11, respectively.

As shown in Table 7 and Fig. 11, the proposed PSTG-ODF model still demonstrates superior recovery performance compared to various baseline methods in real-world power distribution systems.

6) Physical consistency and engineering performance analysis

a) Physical consistency improvement analysis

To quantitatively analyze the effectiveness of the proposed physics-guided mechanism, we examine the degree of violation of physical laws by the imputed values before and after optimization. Taking the IEEE33-node system and recovery scenario two as an example, the optimization objectives involved are shown in ((25)–(26)), and the deviation functions include Φ_1, Φ_2, Φ_5, Φ_6, Φ_7, and Φ_8. For this purpose, we construct six evaluation metrics to measure the degree of violation, with the calculation formula is as follows:

$$Metric1 = \frac{1}{T_*} \sum_{t=1}^{T_*} \sum_{m=1}^{S} \Phi_1(t) \tag{45}$$

where, $\Phi_1(t)$ is the value of the Φ_1 at the t-th missing time section. The calculation formulas for the remaining five evaluation metrics (Metric 2 to Metric 6) are the same as above, i.e., obtained by substituting Φ_2, Φ_5, Φ_6, Φ_7, and Φ_8 into (45), respectively.

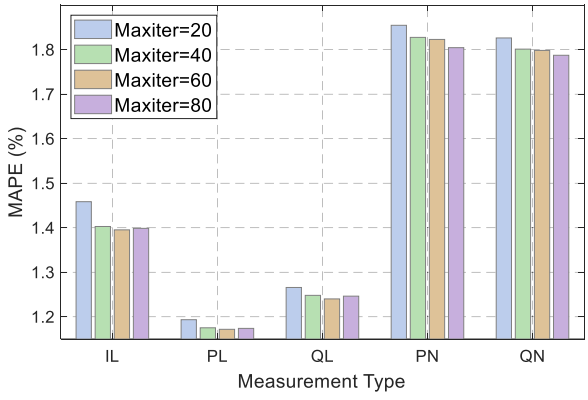


Fig. 8. Recovery MAPE (%) of PSTG-ODF model under different Maxiter values in recovery scenario two ($\tau = 1, \eta = 25\%$).

Table 6
Training time of PSTG-ODF model under different Maxiter values in recovery scenario two ($\tau = 1, \eta = 25\%$).

Maxiter	20	40	60	80
Training time on CPU (min)	203.46	238.93	296.05	340.71

The comparison results of physical deviation before and after preliminary recovery optimization are shown in Fig. 12.

As shown in Fig. 12, all evaluation metrics decrease to varying degrees after preliminary recovery optimization, demonstrating that the physical guidance mechanism effectively reduces the violation of physical laws in the repair results and significantly enhances their physical consistency.

b) Timeliness and resource occupancy analysis

In this subsection, we analyze the time consumption for both offline training and online recovery of various data recovery models. For the IEEE 33-node system, the dataset scale is 323 × 8736, where IL is 96 × 8736, both PL and QL are 64 × 8736, and PN, QN, and UN are each 33 × 8736. The computational time comparisons of the models are shown in Table 8.

As shown in Table 8:

a) During the offline training phase, the training time of the proposed PSTG-ODF model is 270.40 min, which is on the same order of magnitude as several spatio-temporal graph neural network models. Although the introduction of the physics-guided module increases the training time compared to structurally simpler models such as STGCN, a series of simplification strategies, including feasible domain pruning, high-quality initialization, and optimization iteration limits, enable the efficient solution of the optimization models without significantly extending the overall training time.

b) In the online recovery phase, the average recovery time of the PSTG-ODF model for a single time slice is 2.59 s. Given that the sampling intervals for measurement data in distribution systems are typically 15 or 5 min, the recovery time of the proposed model is substantially shorter than the sampling interval, thereby meeting the real-time requirements for data recovery in practical systems.

In addition, to address the requirements for computational resources and real-time reliability in actual DMS deployment, we report in Table 9 the GPU memory usage, CPU memory usage, and iteration count of PSTG-ODF.

The results in Table 9 show that the proposed PSTG-ODF method has a peak GPU memory usage of approximately 1.2GB and a peak CPU memory usage of approximately 2.28GB, indicating that the resource overhead is within a reasonable range. The average number of iterations for training the PSTG-ODF model is 250, with stable and controllable convergence behavior.

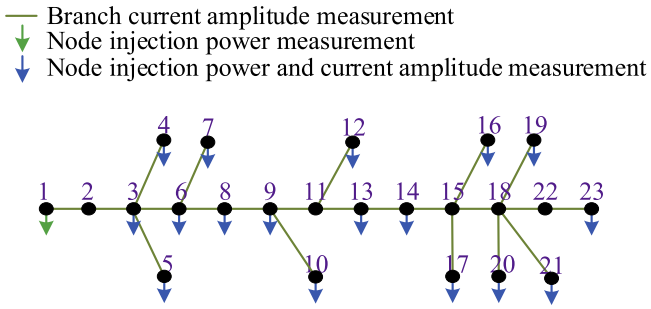


Fig. 9. Measurement configuration of feeder XI46.

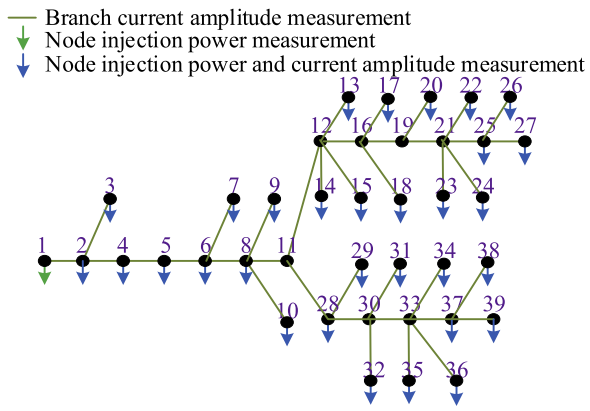


Fig. 10. Measurement configuration of feeder WAN48.

c) Engineering deployment limitations analysis

When the algorithm moves toward practical deployment, it will face many system-level challenges, such as communication delays, dynamic topology changes, and fluctuations in sensor reliability. This section analyzes the limitations of the proposed method in the aforementioned aspects and discusses possible coping strategies.

Communication delays. The single imputation latency of the proposed method is approximately 2–3 s, which is much smaller than the typical sampling period of SCADA systems (5–60 min). When communication delays are within this range, the PSTG-ODF method can function properly; if delays exceed this range, PSTG-ODF will treat them as missing values and perform imputation accordingly.

Dynamic topology changes. The current method assumes that the topology is known and fixed. We recognize that when topology changes occur, the model parameters need to be updated accordingly. In practical deployment, the proposed method can work in conjunction with existing topology identification modules (the module triggers a model update upon detecting a topology change).

Fluctuations in sensor reliability. The proposed method has conducted preliminary explorations in this regard, reducing the influence of unreliable sensors through adaptive adjustment of the transparency mask. In practical deployment, it can be considered for use in coordination with a sensor reliability monitoring module.

5. Conclusion

We present a physics-guided spatio-temporal graph neural network with online data fusion model for coordinated recovery of heterogeneous measurement data in power distribution systems. The model captures spatio-temporal correlations and physical constraints of heterogeneous distribution system measurements while fully utilizing online data. Combined with a physics-guided spatio-temporal graph neural network-based preliminary recovery model and an online data fusion-based secondary recovery model, the physics-guided spatio-temporal graph neural network with online data fusion framework demonstrates excellent recovery accuracy.

Comprehensive case studies on the IEEE 33-node distribution system yield the following key conclusions: (1) The proposed physics-guided spatio-temporal graph neural network with online data fusion model achieves superior performance in coordinated recovery of

Table 7
Recovery performance of different models on the XI46 Feeder ($\tau = 1, \eta = 25\%$).

Restoration quantity	Metric	DCRNN	AGCRN	STGCN	STSGCN	BSVT	Softwalk	SAITS	CSDI	PSTG-ODF
IL	RMSE (A)	2.3512	2.1795	2.3016	2.2418	1.8523	1.9187	1.6283	1.5417	**1.2516**
	MAE (A)	1.4813	1.3688	1.4524	1.4117	1.1615	1.1982	1.0176	0.9624	**0.7813**
	MAPE (%)	5.2487	4.9215	5.1793	5.0482	4.2796	4.4483	3.8172	3.6085	**2.8895**
PN	RMSE (MW)	0.0118	0.0107	0.0115	0.0111	0.0091	0.0095	0.0082	0.0076	**0.0053**
	MAE (MW)	0.0054	0.0049	0.0053	0.0051	0.0042	0.0044	0.0038	0.0035	**0.0024**
	MAPE (%)	7.8185	7.1489	7.6524	7.3816	6.2532	6.4827	5.6234	5.2076	**3.6183**
QN	RMSE (Mvar)	0.0185	0.0168	0.018	0.0173	0.0142	0.0148	0.0126	0.0114	**0.0082**
	MAE (Mvar)	0.0092	0.0084	0.009	0.0086	0.0071	0.0074	0.0063	0.0057	**0.0041**
	MAPE (%)	8.4487	7.7189	8.2516	7.9785	6.8224	7.0789	6.1485	5.6829	**3.9187**

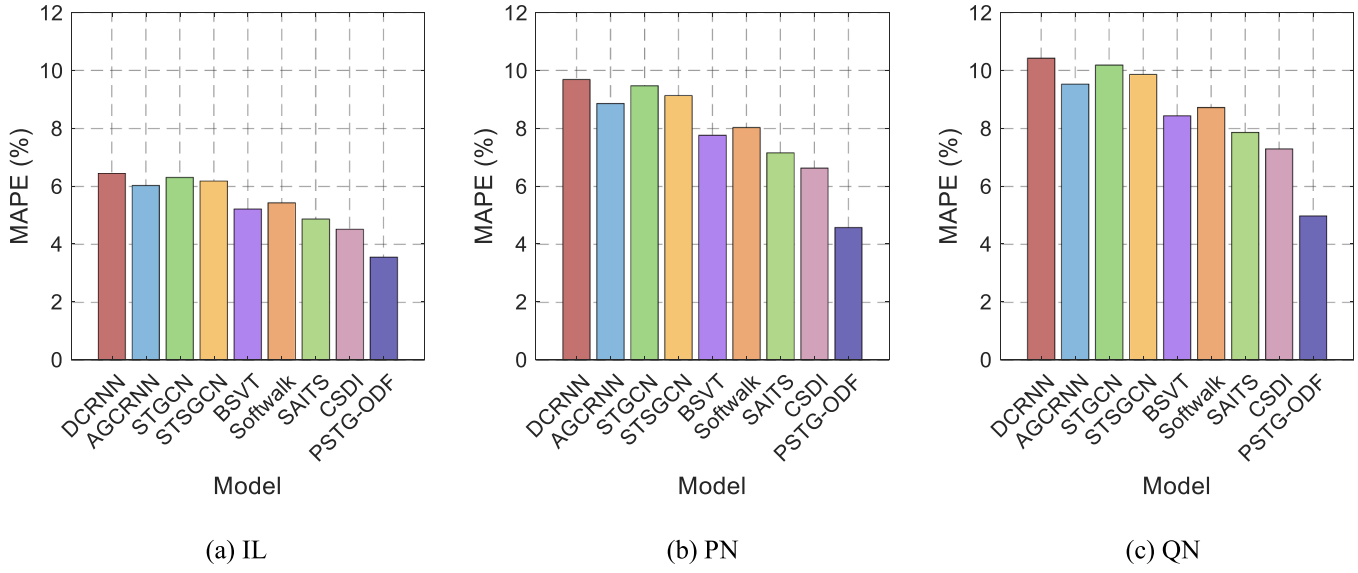


Fig. 11. Recovery MAPE (%) of different models on the WAN48 Feeder ($\tau = 1, \eta = 25\%$).

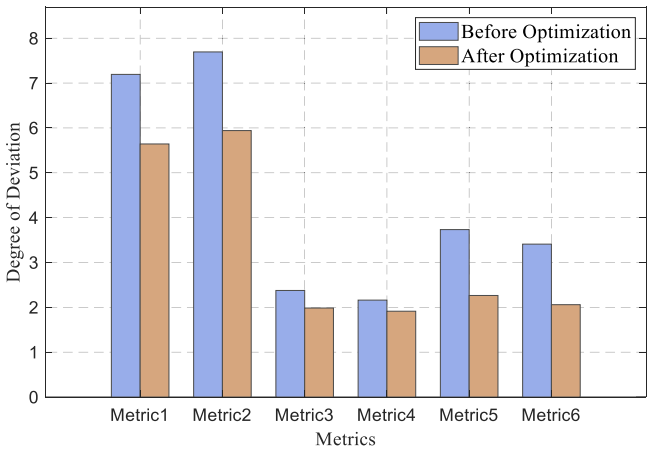


Fig. 12. Comparison of the degree of physical deviation before and after preliminary recovery optimization.

Table 8
Comparison of computational time among different models in the IEEE 33-node system.

Model	Training size	Testing size	Training time on CPU (min)	Online recovery time	
				Total (min)	Per instance (s)
DCRNN	323 × 6989	323 × 1747	252.42	8.44	0.29
AGCRN	323 × 6989	323 × 1747	306.19	12.23	0.42
STGCN	323 × 6989	323 × 1747	216.73	2.33	0.08
STSGCN	323 × 6989	323 × 1747	378.55	21.55	0.74
BSVT	-	323 × 1747	-	2130.47	73.17
Softwalk	323 × 6989	323 × 1747	156.34	48.04	1.65
SAITS	323 × 6989	323 × 1747	185.62	15.38	0.53
CSDI	323 × 6989	323 × 1747	394.56	32.17	1.10
PSTG-ODF	323 × 6989	323 × 1747	270.40	75.41	2.59

Table 9
Resource consumption of PSTG-ODF model in the IEEE 33-node system.

Metric	GPU memory usage (Peak / Average)	CPU memory usage (Peak / Average)	Iteration count
Value	1226MB / 872MB	2.28GB / 1.77GB	250 ± 40

heterogeneous distribution system measurements and can meet the real-time requirements for data recovery; (2) Under various recovery scenarios, different missing period lengths and missing rates, the recovery accuracy of physics-guided spatio-temporal graph neural network with online data fusion model significantly outperforms several baseline methods; (3) Ablation experiments show that the collaborative modeling mechanism of spatio-temporal correlations and physical relationships, as well as the online data fusion mechanism, can substantially improve recovery accuracy; (4) The physics-guided spatio-temporal graph neural network with online data fusion model can meet the real-time requirements of the system for data recovery, and maintains good recovery performance on actual systems, demonstrating excellent generalizability and generalization capability.

In terms of application value, this study explores an artificial intelligence approach that integrates physics-guided and data-driven methods to address the urgent demand for high-quality measurement data in core operations of new electricity systems, such as renewable energy integration and virtual power plant cooperative control. The proposed physics-guided spatio-temporal graph neural network with online data fusion model can be embedded into the data preprocessing module of distribution system dispatch centers to perform online recovery of missing measurement data, thereby providing high-quality support for advanced applications such as state estimation and fault diagnosis. The model adopts a lightweight design, with single recovery latency meeting real-time requirements, demonstrating potential for deployment in distribution network dispatch systems. This research offers insights with potential implications for advancing the application of artificial intelligence in the energy sector.

CRediT authorship contribution statement

Wei Wei: Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Formal analysis. **Pan Huang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Tao Xu:** Writing – review & editing, Supervision, Project administration, Investigation, Formal analysis. **Lingxu Guo:** Supervision, Resources, Project administration, Funding acquisition. **Xu Huang:** Supervision, Resources, Data curation. **Yang Wang:** Resources, Investigation, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

The formulas for credibility assessment indicators are as follows:

$$\zeta_1 = E_{\mathrm{rr}} \times 100\% \tag{A.1}$$

$$\zeta_2 = \frac{N_{\mathrm{fail}}}{N_{\mathrm{run}}} \times 100\% \tag{A.2}$$

$$\zeta_3 = 0.01 \times \frac{N_{\mathrm{day}}}{365} \times 100\% \tag{A.3}$$

$$\zeta_4 = \begin{cases} 0.9 \sim 1.0 & \text{Level1} \\ 0.7 \sim 0.9 & \text{Level2} \\ 0.5 \sim 0.7 & \text{Level3} \\ 0 \sim 0.5 & \text{Level4} \end{cases} \tag{A.4}$$

$$\zeta_5 = \frac{N_{\text{missing}}}{N_{\text{total}}} \times 100\% \tag{A.5}$$

$$\zeta_6 = \frac{|V_{\text{meas}} - V_{\text{theor}}|}{V_{\text{dev}}} \times 100\% \tag{A.6}$$

where E_{rr} represents the manufacturer-provided nominal error of the device, N_{fail} denotes failure duration in hours, N_{run} indicates total operation hours, N_{day} stands for device service days, N_{missing} is the number of missing data points in the statistical period, N_{total} is the total theoretically collectable data points, V_{meas} stands for measured values, V_{theor} represents theoretical values (power flow results, state estimation results, etc.), V_{dev} indicates the allowable deviation threshold, Level 1 represents ideal conditions, Level 2 indicates mildly harsh conditions, Level 3 denotes moderately harsh conditions, Level 4 signifies severely harsh conditions.

Appendix B

The calculation formulas for the deterministic component $\mathbb{F}_\alpha(\Re_\beta)$ and the uncertain component $\mathbb{F}_\alpha(\Re_0)$ of the basic belief assignment function $\mathbb{F}_\alpha(\cdot)$ are as follows:

$$\mathbb{F}_\alpha(\Re_\beta) = \frac{\mu_{(\alpha,\beta)}}{\sum\limits_{\alpha,\beta}\mu_{(\alpha,\beta)} + \sum\limits_{\alpha}\mu_\alpha} \tag{B.1}$$

$$\mathbb{F}_\alpha(\Re_0) = \frac{\mu_\alpha}{\sum\limits_{\alpha,\beta}\mu_{(\alpha,\beta)} + \sum\limits_{\alpha}\mu_\alpha} \tag{B.2}$$

where $\alpha = \zeta_1, \zeta_2, \zeta_3, \zeta_4, \zeta_5, \zeta_6$, $\beta = 1, 2, 3, 4$, $\mu_{(\alpha,\beta)}$ denotes the membership function, μ_α represents the uncertainty function.

The membership function for measurement device error ζ_1 is formulated as follows, and the same applies to others.

$$\mu_{(\zeta_1,1)} = \begin{cases} 1 & \zeta_1 \le 0.02 \\ 0 & \zeta_1 > 0.02 \end{cases} \tag{B.3}$$

$$\mu_{(\zeta_1,2)} = \begin{cases} \dfrac{\zeta_1 - 0.01}{0.02 - 0.01} & 0.01 < \zeta_1 \le 0.02 \\ 1 & 0.02 < \zeta_1 \le 0.04 \\ e^{-100(\zeta_1 - 0.04)} & 0.04 < \zeta_1 < 0.06 \\ 0 & \zeta_1 \le 0.01, \zeta_1 \ge 0.06 \end{cases} \tag{B.4}$$

$$\mu_{(\zeta_1,3)} = \begin{cases} e^{-100(0.06 - \zeta_1)} & 0.04 < \zeta_1 \le 0.06 \\ e^{-20(\zeta_1 - 0.06)} & 0.06 < \zeta_1 < 0.1 \\ 0 & \zeta_1 \le 0.04, \zeta_1 \ge 0.1 \end{cases} \tag{B.5}$$

$$\mu_{(\zeta_1,4)} = \begin{cases} 1 & \zeta_1 \ge 0.1 \\ \dfrac{\zeta_1 - 0.06}{0.1 - 0.06} & 0.06 < \zeta_1 < 0.1 \\ 0 & \zeta_1 \le 0.06 \end{cases} \tag{B.6}$$

The uncertainty function μ_α is formulated as:

$$\mu_\alpha = 1 - \max\left(\mu_{(\alpha,1)}, \mu_{(\alpha,2)}, \mu_{(\alpha,3)}, \mu_{(\alpha,4)}\right) \tag{B.7}$$

where $\alpha = \zeta_1, \zeta_2, \zeta_3, \zeta_4, \zeta_5, \zeta_6$.

For all credibility levels, the final belief degree $\mathbb{F}(\Re_\beta)$ is formulated as:

$$\mathbb{F}(\Re_\beta) = \left(\mathbb{F}_{\zeta_1} \oplus \mathbb{F}_{\zeta_2} \oplus \mathbb{F}_{\zeta_3} \oplus \mathbb{F}_{\zeta_4} \oplus \mathbb{F}_{\zeta_5} \oplus \mathbb{F}_{\zeta_6}\right)(\Re_\beta) = \frac{\sum\limits_{\delta^{(\zeta_1)} \cap \delta^{(\zeta_2)} \cap \ldots \cap \delta^{(\zeta_6)} = \Re_\beta} \left(\prod\limits_{\alpha} \mathbb{F}_\alpha(\delta^{(\alpha)})\right)}{1 - \sum\limits_{\delta^{(\zeta_1)} \cap \delta^{(\zeta_2)} \cap \ldots \cap \delta^{(\zeta_6)} = \phi} \left(\prod\limits_{\alpha} \mathbb{F}_\alpha(\delta^{(\alpha)})\right)} \tag{B.8}$$

where $\delta^{(\alpha)}$ denotes $\Re_1$, $\Re_2$, $\Re_3$, $\Re_4$ or $\Re_0$ corresponding to indicator α, ϕ is the empty set, $\beta = 1, 2, 3, 4$.

Appendix C

a) Analysis of the nature of the optimization problem

The pre-repair optimization objective function ((23)–(28)) constructed by the physics-guided module in this paper has the following form:

$$\min_{x} \frac{\| Ax + b \|_2}{c^{\mathrm{T}}x + d} \tag{C.1}$$

The numerator is a convex function (norms are convex functions), and the denominator is a linear function. From a functional perspective, it is neither convex nor concave. However, it can be transformed into an equivalent convex optimization problem through variable transformation. The specific steps are as follows:

Let:

$$y = \frac{x}{c^{\mathrm{T}}x + d},\ t = \frac{1}{c^{\mathrm{T}}x + d} \tag{C.2}$$

Then $x = \frac{y}{t}$, and $c^{\mathrm{T}}x + d = \frac{c^{\mathrm{T}}y}{t} + d = \frac{c^{\mathrm{T}}y + dt}{t} = \frac{1}{t}$, from which it follows that:

$$c^{\mathrm{T}}y + dt = 1 \tag{C.3}$$

The original objective is transformed into:

$$\frac{\| Ax + b \|_2}{c^{\mathrm{T}}x + d} = \frac{\| A\frac{y}{t} + b \|_2}{1/t} = t\| A\frac{y}{t} + b \|_2 = \| Ay + bt \|_2 \tag{C.4}$$

This yields a second-order conic program:

$$\begin{aligned} &\min_{y,t} \| Ay + bt \|_2 \\ &\text{s.t.} c^{\mathrm{T}}y + dt = 1, t > 0 \end{aligned} \tag{C.5}$$

The objective is to minimize the norm, and the constraints are linear, which constitutes a convex optimization problem with a globally optimal solution.

The secondary recovery optimization objectives constructed in the ODF stage ((36)–(41)) are obtained by adding a weighted penalty term to the preliminary recovery optimization objective function. The weighted penalty term is determined by (35) and takes the form of a norm, which is also a convex function. Therefore, the overall convexity of the problem is maintained.

b) Soft constraints and limited iteration strategy

For the optimization problems in the preliminary recovery and secondary recovery stages in this paper, the constraints are determined by (29), which are linear inequality constraints and do not alter the convexity of the optimization problem.

From the above analysis, it can be concluded that the optimization problem is a convex quadratic programming problem. For such problems, the SLSQP solver generates a descent direction at each iteration by solving a quadratic programming subproblem and determines the step size using a line search, ensuring that the objective function value is strictly non-increasing. Therefore, the iterative process consistently moves toward the optimal solution. Moreover, the initial point is the output value of the neural network, which provides a high-quality starting point for the solver, enabling fast convergence and achieving high-precision solutions within a limited number of iterations.

Appendix D

In this section, we conduct a sensitivity analysis on the number of credibility levels.

Taking the IEEE 33-node distribution system and recovery scenario two as an example, we set three types of numbers for the credibility levels: two levels, four levels, and six levels. The recovery performance and online recovery time of the PSTG-ODF model under missing period length $\tau = 1$ and missing rate $\eta = 25\%$ are shown in Tables D.1 and D.2, respectively. The relevant settings for "four levels" are presented in Section 3.3. Based on this, the different settings for "two levels" are given in Tables D.3 and Eq.(D.1)– Eq.(D.4), while the different settings for "six levels" are given in Table D.4 and Eq.(D.5)–Eq.(D.12).

Table D.1
Effects of the number of credibility levels in recovery scenario two ($\tau = 1$,$\eta = 25\%$).

Restoration quantity	Metric	Two levels	Four levels	Six levels
IL	MAPE (%)	1.6528	1.4028	1.3556
PL		1.3986	1.1752	1.0895
QL		1.4853	1.2481	1.1702
PN		2.2756	1.8276	1.7138
QN		2.2432	1.8012	1.6998

Table D.2
Online recovery time of the PSTG-ODF model in recovery scenario two ($\tau = 1, \eta = 25\%$).

Model	Testing size	Online recovery time	Two levels	Four levels	Six levels
PSTG-ODF	290 × 1747	Total (min)	57.45	67.55	85.60
		Per instance (s)	1.87	2.32	2.94

As shown in Table D.1, the finer the credibility classification, the higher the data recovery accuracy, with the MAPE value showing an overall downward trend. The improvement from two levels to four levels is relatively large, while the improvement from four levels to six levels is smaller, indicating that the gain from refining the classification from coarse-grained to medium-grained is the most significant, whereas further refinement yields diminishing returns. Moreover, as shown in Table D.2, the online recovery time also increases with the number of credibility levels. Overall, the four-level credibility classification achieves a good balance between recovery performance and computational efficiency, making it a more desirable choice.

a) Relevant settings for "two levels"

The measurement credibility is divided into two levels: credible $\Re_1$ and non-credible $\Re_2$, as shown in Table D.3.

Table D.3
Measurement credibility levels.

Evaluation indicators	Credibility level	
	$\Re_1$	$\Re_2$
ζ_1	[0, 6%)	(4%, 1]
ζ_2	[0, 25%)	(20%, 1]
ζ_3	[0, 4%)	(3%, 1]
ζ_4	(0.6, 1]	[0, 0.7)
ζ_5	[0, 25%)	(20%, 1]
ζ_6	[0, 25%)	(20%, 1]

The membership function for measurement device error ζ_1 is formulated as follows, and the same applies to others.

$$\mu_{(\zeta_1,1)} = \begin{cases} 1 & \zeta_1 \le 0.04 \\ \dfrac{0.06 - \zeta_1}{0.06 - 0.04} & 0.04 < \zeta_1 < 0.06 \\ 0 & \zeta_1 \le 0.04 \end{cases} \tag{D.1}$$

$$\mu_{(\zeta_1,2)} = \begin{cases} 1 & \zeta_1 \ge 0.06 \\ \dfrac{\zeta_1 - 0.04}{0.06 - 0.04} & 0.04 < \zeta_1 < 0.06 \\ 0 & \zeta_1 \le 0.04 \end{cases} \tag{D.2}$$

The calculation formulas for the transparency value ϑ are as follows:

$$\mathbb{F}_{\max} = \max(\mathbb{F}(\Re_1), \mathbb{F}(\Re_2)) \tag{D.3}$$

$$\vartheta = \begin{cases} 0.9 & \text{if} \mathbb{F}(\Re_1) = \mathbb{F}_{\max} \\ 0 & \text{if} \mathbb{F}(\Re_2) = \mathbb{F}_{\max} \end{cases} \tag{D.4}$$

b) Relevant settings for "six levels"

The measurement credibility is divided into six levels: highly credible $\Re_1$, credible $\Re_2$, moderately credible $\Re_3$, basically credible $\Re_4$, less credible $\Re_5$, and non-credible $\Re_6$, as shown in Table D.4.

Table D.4
Measurement credibility levels.

Evaluation indicators	Credibility level					
	$\Re_1$	$\Re_2$	$\Re_3$	$\Re_4$	$\Re_5$	$\Re_6$
ζ_1	[0, 2%)	(1%, 6%)	(4%, 10%)	(8%, 20%)	(15%, 25%)	[25%, 1]
ζ_2	[0, 15%)	(10%, 25%)	(20%, 40%)	(30%, 50%)	(45%, 60%)	[60%, 1]
ζ_3	[0, 2%)	(1%, 4%)	(3%, 6%)	(5%, 10%)	(8%, 12%)	[12%, 1]
ζ_4	(0.8, 1]	(0.6, 0.9)	(0.4, 0.7)	(0.2, 0.5)	(0.15, 0.3)	[0,0.3]
ζ_5	[0, 15%)	(10%, 25%)	(20%, 40%)	(30%, 50%)	(45%, 60%)	[60%, 1]
ζ_6	[0, 15%)	(10%, 25%)	(20%, 40%)	(30%, 50%)	(45%, 60%)	[60%, 1]

The membership function for measurement device error ζ_1 is formulated as follows, and the same applies to others.

$$\mu_{(\zeta_1,1)} = \begin{cases} 1 & \zeta_1 \le 0.02 \\ 0 & \zeta_1 > 0.02 \end{cases} \tag{D.5}$$

$$\mu_{(\zeta_1,2)} = \begin{cases} \dfrac{\zeta_1 - 0.01}{0.02 - 0.01} & 0.01 < \zeta_1 \le 0.02 \\ 1 & 0.02 < \zeta_1 \le 0.04 \\ e^{-100(\zeta_1 - 0.04)} & 0.04 < \zeta_1 < 0.06 \\ 0 & \zeta_1 \le 0.01, \zeta_1 \ge 0.06 \end{cases} \tag{D.6}$$

$$\mu_{(\zeta_1,3)} = \begin{cases} \dfrac{\zeta_1 - 0.04}{0.06 - 0.04} & 0.04 < \zeta_1 \le 0.06 \\ 1 & 0.06 < \zeta_1 \le 0.08 \\ e^{-100(\zeta_1 - 0.08)} & 0.08 < \zeta_1 < 0.1 \\ 0 & \zeta_1 \le 0.04, \zeta_1 \ge 0.1 \end{cases} \tag{D.7}$$

$$\mu_{(\zeta_1,4)} = \begin{cases} \dfrac{\zeta_1 - 0.08}{0.02 - 0.08} & 0.08 < \zeta_1 \le 0.1 \\ 1 & 0.1 < \zeta_1 \le 0.15 \\ e^{-100(\zeta_1 - 0.15)} & 0.15 < \zeta_1 < 0.2 \\ 0 & \zeta_1 \le 0.08, \zeta_1 \ge 0.2 \end{cases} \tag{D.8}$$

$$\mu_{(\zeta_1,5)} = \begin{cases} e^{-100(0.2 - \zeta_1)} & 0.15 < \zeta_1 \le 0.2 \\ e^{-20(\zeta_1 - 0.2)} & 0.2 < \zeta_1 < 0.25 \\ 0 & \zeta_1 \le 0.15, \zeta_1 \ge 0.2 \end{cases} \tag{D.9}$$

$$\mu_{(\zeta_1,6)} = \begin{cases} 1 & \zeta_1 \ge 0.25 \\ \dfrac{\zeta_1 - 0.2}{0.25 - 0.2} & 0.2 < \zeta_1 < 0.25 \\ 0 & \zeta_1 \le 0.2 \end{cases} \tag{D.10}$$

The calculation formulas for the transparency value ϑ are as follows:

$$\mathbb{F}_{\max} = \max(\mathbb{F}(\Re_1), \mathbb{F}(\Re_2), \mathbb{F}(\Re_3), \mathbb{F}(\Re_4), \mathbb{F}(\Re_5), \mathbb{F}(\Re_6)) \tag{D.11}$$

$$\vartheta = \begin{cases} 0.9 & \text{if} \mathbb{F}(\Re_1) = \mathbb{F}_{\max} \\ 0.7 & \text{if} \mathbb{F}(\Re_2) = \mathbb{F}_{\max} \\ 0.5 & \text{if} \mathbb{F}(\Re_3) = \mathbb{F}_{\max} \\ 0.3 & \text{if} \mathbb{F}(\Re_4) = \mathbb{F}_{\max} \\ 0.1 & \text{if} \mathbb{F}(\Re_5) = \mathbb{F}_{\max} \\ 0 & \text{if} \mathbb{F}(\Re_6) = \mathbb{F}_{\max} \end{cases} \tag{D.12}$$

Data availability

The data that has been used is confidential.

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